

pi0eta_ampmodel_fit: an amplitude-level model of π^0 and η electroproduction fitted to CLAS, Hall A and COMPASS

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5 October 2026 – fit Mom_B12

Abstract

A model of exclusive π^0 and η electroproduction is built directly at the level of the helicity amplitudes, without reference to generalised parton distributions, and fitted to 963 measured points from CLAS6, CLAS12, Hall A and COMPASS. The purpose is not to test a particular GPD parametrisation but to establish what the data themselves demand of the amplitudes, so that any model can be measured against it. The fit has 30 free parameters and gives $\chi^2/\text{ndf} = 1.6868$ (1573.8/933).

Three things changed since the September version of this report, and each changed an answer rather than the bookkeeping. A $\langle \tilde{H} \rangle$ block was added, because without it σ_L vanished at the forward peak against the model's own angular-momentum counting. The Hall A Rosenbluth-separated proton σ_L and the neutron σ_U at two beam energies were recovered from the publications: neither had ever reached the workbook, which has no column for a separated σ_L . And the De Masi beam-spin asymmetry now enters as 60 moments extracted from its ϕ distributions rather than as 663 ϕ points, which had been supplying 42% of the fit while carrying about sixty independent numbers.

Four conclusions survive. The n/p flavour ratio is soft against a change of model assumption in a way no error bar expresses. One polarised-target experiment alone fixes the relative phase of two transverse amplitudes. The *sign* of σ_{LT} is set by a phase difference, not by a modulus, and three of the seven Hall A 2016 points are described by no smooth model. And a single power Q^{n_Q} cannot span Q^2 from 1.2 to 8.3 GeV².

1 The model

1.1 What is parametrised

It is worth being explicit about the level at which this model lives, because three different objects are easily confused.

1. The *generalised parton distributions* themselves, $X^q(x, \xi, t)$ for $X \in \{H_T, \bar{E}_T, \dots\}$ and flavour $q \in \{u, d\}$. These depend on the loop momentum fraction x .
2. The *convolution* of a GPD with the hard subprocess, obtained by integrating x out.
3. The *channel helicity amplitudes*, obtained from the convolutions by combining flavours with the photon couplings and the meson wavefunction.

This model never descends to level 1. No integral over x is performed anywhere, and no GPD is assumed. What is parametrised is the convolution, level 2. That is the whole point of the approach: the data are asked what they require of the convolutions, without committing to a GPD parametrisation that produces them.

The convolution is

$$\langle X \rangle^q(\xi, t, Q^2) = \int_{-1}^1 dx X^q(x, \xi, t) K(x, \xi), \quad K(x, \xi) \sim \frac{1}{x - \xi + i\epsilon}, \quad (1)$$

where K is the hard-scattering kernel. Its precise form is not needed below, because (1) is never evaluated – only parametrised. What matters is a structural consequence of the $i\epsilon$: the convolution is a *complex number*,

$$\text{Im}\langle X \rangle^q = \pi X^q(\xi, \xi, t), \quad \text{Re}\langle X \rangle^q = \mathcal{P} \int dx X^q K, \quad (2)$$

the imaginary part coming from the pole and the real part from the principal value. Since the x -shapes of u and d differ, $\langle X \rangle^u$ and $\langle X \rangle^d$ cannot be assumed to share a phase.

The ansatz. Each convolution is parametrised by a modulus and a phase, separately for each flavour:

$$|\langle X \rangle^q| = N_q \exp[(b_q + b'_q \ln x_B) t] (Q^2)^{n_Q^q/2}, \quad (3)$$

$$\arg\langle X \rangle^d - \arg\langle X \rangle^u = \varphi_X. \quad (4)$$

The overall phase of a channel is unobservable, so u is taken real and only d carries φ_X . The parameters mean:

N	normalisation
b	t -slope extrapolated to $x_B = 1$
b'	the slope at a given x_B is $b + b' \ln x_B$, so $-b'$ plays the role of the Regge α'
n_Q	power of Q^2
φ	phase of the d convolution relative to u

This is the same slope convention as the Goloskokov–Kroll fits. An older parameter convention used $L = \ln x_B - \ln 0.15$, so that b was the slope at $x_B = 0.15$; files in that convention must be converted ($b_{\text{new}} = b + 1.897 b'$), never copied raw.

Three blocks are parametrised this way: $\langle H_T \rangle$ and $\langle \bar{E}_T \rangle$, which build the transverse amplitudes, and a longitudinal block $\langle L \rangle$, which builds $T_{00}^{(-)}$. The labels H_T and $\bar{E}_T$ name the GPD being convolved; the object that carries the parameters is always the convolution, never the GPD itself.

Assumptions, stated plainly. Equations (3) and (4) embody two approximations that should not be hidden. A single exponential in t for the modulus, and a t -independent phase, are together equivalent to assuming that $\text{Re}\langle X \rangle$ and $\text{Im}\langle X \rangle$ share a common t -slope: if they did not, the modulus $\sqrt{\text{Re}^2 + \text{Im}^2}$ would not be a single exponential and $\varphi = \arctan(\text{Im} / \text{Re})$ would depend on t . Nothing forces that common slope; it is an assumption of convenience, and relaxing it costs one parameter per block ($\varphi \rightarrow \varphi_0 + \varphi_1 t$, as already used for the phase of $T_{00}^{(-)}$).

1.2 From convolutions to channel amplitudes

The observables are built from six channel helicity amplitudes. In the notation of the π^0 amplitude note,

$T_{00}^{(+)}$	longitudinal, recoil non-flip	$T_{00}^{(-)}$	longitudinal, recoil flip
$T_{01}^{(+)}$	natural transverse, non-flip	$T_{01}^{(-)}$	natural transverse, flip
$U_{01}^{(+)}$	unnatural transverse, non-flip	$U_{01}^{(-)}$	unnatural transverse, flip

At the level used here $T_{01}^{(-)} = U_{01}^{(-)} = D/2$, real and equal, which is the forward-kinematics relation; the remaining four carry the fitted moduli and phases.

The photon couples to quark q with charge e_q , and the meson wavefunction selects which combination survives. With $\pi^0 = (u\bar{u} - d\bar{d})/\sqrt{2}$ and $\eta_8 = (u\bar{u} + d\bar{d} - 2s\bar{s})/\sqrt{6}$, the coefficient of each convolution is $\sum_q (\text{meson amplitude}) \times e_q$. The strange term drops out, for the reason given below. Writing $X_q \equiv |\langle X \rangle^q|$, this gives

$$\langle X \rangle_{\pi^0 p} = \frac{2X_u + X_d e^{i\varphi_X}}{3\sqrt{2}}, \quad \langle X \rangle_{\eta p} = \frac{2X_u - X_d e^{i\varphi_X}}{3\sqrt{6}k}, \quad \langle X \rangle_{\pi^0 n} = \frac{X_u + 2X_d e^{i\varphi_X}}{3\sqrt{2}}. \quad (5)$$

A superscript on $\langle X \rangle$ is a flavour, a subscript is a channel: $\langle X \rangle^u$ and $\langle X \rangle^d$ are the convolutions of (1), while $\langle X \rangle_{\text{ch}}$ is the channel combination built from them. The relative sign differs between π^0 and η because $e_d < 0$ meets $-d\bar{d}$ in the π^0 and $+d\bar{d}$ in the η . The neutron entry is not new physics: it is the same meson factor with isospin applied to the *target*, $X_u^n = X_d^p$ and $X_d^n = X_u^p$.

Why there is no strange term, and no gluon term. Pseudoscalar meson production filters the C -odd, valence-like combination of quark GPDs – the hard kernel enters antisymmetrised in $x \rightarrow -x$, so quark and antiquark contributions appear with opposite sign. A symmetric sea, $q_{\text{sea}} = \bar{q}_{\text{sea}}$, therefore cancels identically in that combination. Strangeness in the nucleon is pure sea, so $\langle X \rangle^s = 0$ to the extent that $s = \bar{s}$; the approximation fails only through the strangeness asymmetry $s - \bar{s}$, which is small. This is why the $-2s\bar{s}$ piece of η_8 can be dropped in (5) without an additional parameter.

The same C -parity argument removes the gluon contribution altogether, which is what makes π^0 and η production a comparatively clean probe of the valence sector: unlike vector mesons, there is no gluon GPD entering at this order. The singlet component of the η survives only through the η - η' mixing of the decay constants, which is carried by k_L in (7), not by an independent amplitude.

The combination in (5) is a sum of *amplitudes*; the squaring happens only later, in (10). The phase φ_X is therefore physical and not removable: at $\varphi_X = 0$ the η amplitude can pass through an exact zero when $X_d = 2X_u$, whereas for $\varphi_X \neq 0$

$$|2X_u - X_d e^{i\varphi_X}|^2 = 4X_u^2 + X_d^2 - 4X_u X_d \cos \varphi_X \quad (6)$$

has a floor of $4X_u^2 \sin^2 \varphi_X$. Note also that this expression is *even* in φ_X : the unpolarised structure functions cannot determine the sign of the phase, which is fixed only by the interference terms σ_{LT} and $\sigma_{LT'}$.

The factor k is not flavour algebra but meson normalisation, and it enters as a divisor, so $1/k$ is an enhancement of the η channel. In the longitudinal twist-2 sector the meson enters through its decay constant; in the two-mixing-angle scheme ($\theta_8 = -21.2^\circ$, $\theta_1 = -9.2^\circ$, $f_8 = 1.26f_\pi$, $f_1 = 1.17f_\pi$),

$$k_L^{-1} = \left(\cos \theta_8 - \sqrt{2} \frac{f_1}{f_8} \sin \theta_1 \right) \frac{f_8}{f_\pi} = 1.439 \quad \implies \quad k_L = 0.695. \quad (7)$$

In the transverse twist-3 sector the meson enters through the chiral-odd twist-3 distribution amplitude, normalised by $\mu_M \sim m_M^2/(m_u + m_d)$, giving $k_T = 0.863$.

The near cancellation $2X_u - X_d$ in the η channel is what makes that channel sensitive to the flavour separation, and it is the reason η data carry weight far beyond their number.

The amplitudes themselves. Writing $\xi = \frac{x_B}{2-x_B} (1 + M_p^2/Q^2)$, $t' = t - t_{\min}$ and $A = A(Q^2, x_B)$ for the phase-space factor,

$$D = \sqrt{2A(1-\xi^2)} \langle H_T \rangle_{\text{ch}}, \quad U_{01}^{(+)} = \sqrt{A\kappa} \langle \bar{E}_T \rangle_{\text{ch}}, \quad (8)$$

with $\kappa = -t'/(8M_p^2)$ and the subscript “ch” the channel combination of (5). Three parameters unlock what would otherwise be exact relations: ρ_{CE} and ϕ_{CE} give $T_{01}^{(+)}$ a modulus and phase relative to $U_{01}^{(+)}$, breaking $C = E$, and ϕ_w is the phase of $U_{01}^{(+)}$.

The longitudinal sector, corrected 30 September 2026. Until that date both longitudinal amplitudes carried $\sqrt{-t'}$, so $\sigma_L \propto -t' \rightarrow 0$ at the forward peak. That contradicts the angular-momentum counting of the amplitude note: with $\Delta = \mu - \nu + \nu'$ the amplitude goes as $(\sqrt{-t'})^{|\Delta|}$, and the non-flip $M_{0+,0+}$ has $\Delta = 0$, so it must be *constant* at $t' = 0$, not vanishing. What was missing is $\langle \tilde{H} \rangle$. In the GK dictionary

$$M_{0+,0+} \sim \sqrt{1 - \xi^2} \left[\langle \tilde{H} \rangle - \frac{\xi^2}{1 - \xi^2} \langle \tilde{E} \rangle \right], \quad M_{0-,0+} \sim \frac{\sqrt{-t'}}{2M_p} \xi \langle \tilde{E} \rangle, \quad (9)$$

so the old longitudinal block is re-read as $\langle \tilde{E} \rangle$ and $\langle \tilde{H} \rangle$ is new: four parameters N, b, b', n_Q and one d/u ratio, with the u - d relative phase shared with $\langle \tilde{E} \rangle$.

Two guards were needed and are part of the model. First, $\xi \geq 1$ is outside GPD support and the $\xi^2/(1 - \xi^2)$ pole is not a physical enhancement: a point with $\xi \geq \xi_{\max} = 0.8$ is rejected rather than clamped. All fitted data have $\xi \leq 0.486$. Second, the shape of $\langle \tilde{E} \rangle$ is *tied* to that of $\langle \tilde{H} \rangle$ — both b' and n_Q — leaving it its own normalisation and d/u ratio. Left free it drifted where no data holds it: n_Q turned negative, so the block grew as $Q^2 \rightarrow 0$, and the normalisation rose fourfold. Neither is visible in χ^2 , because in the fitted region the $\langle \tilde{E} \rangle$ piece of $T_{00}^{(+)}$ is 1.5 against $\langle \tilde{H} \rangle$'s 3.8; but the radiative integral of the generator reaches $Q_t^2 = 0.084$, where the same piece is 15990 against 176, and the correction factor came out 10^9 instead of 10. A fit cannot see a region it has no data in, and a generator must survive it.

The phase of $\langle \tilde{E} \rangle$ is $\delta_0 + \delta_1 t$, common to both flavours — this one phase is allowed a t -dependence, unlike φ_X — and $\langle \tilde{H} \rangle$ carries its own constant phase ϕ_{nf} . Section 5 shows that those two phases, and not the moduli, decide the *sign* of σ_{LT} .

1.3 What is free and what is tied

The flavours are not treated symmetrically, and the pattern is a model assumption worth stating. For fit Mom_B12:

block	slope u [GeV ⁻² , at $x_B = 0.25$]	slope d	d/u	phase φ [rad]
$\langle H_T \rangle$	1.810	1.810 (tied)	-0.096	-0.429
$\langle \bar{E}_T \rangle$	1.254	5.666 (free)	+2.777	+0.454
$\langle \tilde{E} \rangle$	0.921	= u	-0.802	+0.954
$\langle \tilde{H} \rangle$	0.921	= u	-0.682	+2.729

Only $\langle \bar{E}_T \rangle$ is allowed an independent t -slope per flavour, and it uses that freedom: the d slope comes out four and a half times the u slope. Everywhere else the two flavours share one slope and differ only in normalisation and phase, and the two longitudinal blocks share their whole shape — b' and n_Q — by the tie of paragraph 1.2. Whether $\langle H_T \rangle$ would also want a split slope is not tested here.

1.4 From amplitudes to structure functions

The nine structure functions follow from the Fourier dictionary, verified in the note against the QED photon density matrix:

$$\begin{aligned} \sigma_L &= \sum |T_{00}|^2, & \sigma_T &= \sum |T_{01}|^2 + \sum |U_{01}|^2, & \sigma_{TT} &= \sum |T_{01}|^2 - \sum |U_{01}|^2, \\ \sigma_{LT} + i\sigma_{LT'} &= -\sqrt{2} \sum T_{00}^* T_{01}, & \sigma_{LT'}^{LL} + i\sigma_{LT}^{UL} &= -\sqrt{2} \sum T_{00}^* U_{01}, & \sigma_{T'} + i\sigma_{TT}^{UL} &= 2 \sum T_{01}^* U_{01}. \end{aligned} \quad (10)$$

Only the moduli of T_{01} and U_{01} enter σ_T and σ_{TT} ; their *relative phase* appears only in $\sigma_{T'}$. This observation matters later: it is why one polarised-target experiment is the sole constraint on that phase.

The measured cross section, for beam helicity $h = \pm 1$ and a *longitudinally* polarised target of polarisation Λ , is

$$\begin{aligned} \frac{d^2\sigma}{dt d\phi} = \frac{1}{2\pi} & \left[\sigma_T + \epsilon \sigma_L + \epsilon \cos 2\phi \sigma_{TT} + \sqrt{2\epsilon(1+\epsilon)} \cos \phi \sigma_{LT} + h \sqrt{2\epsilon(1-\epsilon)} \sin \phi \sigma_{LT'} \right. \\ & + \Lambda \left(\sqrt{2\epsilon(1+\epsilon)} \sin \phi \sigma_{LT}^{UL} + \epsilon \sin 2\phi \sigma_{TT}^{UL} \right) \\ & \left. + h \Lambda \left(\sqrt{1-\epsilon^2} \sigma_{T'} + \sqrt{2\epsilon(1-\epsilon)} \cos \phi \sigma_{LT'}^{LL} \right) \right], \end{aligned} \quad (11)$$

with ϕ in the Trento convention. The first line, which is all that survives at $\Lambda = 0$, is identical to eq. (B4) of the CLAS6 paper and eq. (2) of COMPASS. The second line is the target-spin asymmetry A_{UL} and the third the double-spin asymmetry A_{LL} ; dividing each term by $\sigma_0 = \sigma_T + \epsilon \sigma_L$ gives the four moments that are measured and that enter this fit,

$$A_{UL}^{\sin \phi} = \frac{\sqrt{2\epsilon(1+\epsilon)} \sigma_{LT}^{UL}}{\sigma_0}, \quad A_{UL}^{\sin 2\phi} = \frac{\epsilon \sigma_{TT}^{UL}}{\sigma_0}, \quad A_{LL}^{\text{const}} = \frac{\sqrt{1-\epsilon^2} \sigma_{T'}}{\sigma_0}, \quad A_{LL}^{\cos \phi} = \frac{\sqrt{2\epsilon(1-\epsilon)} \sigma_{LT'}^{LL}}{\sigma_0}. \quad (12)$$

Two remarks on the scope of eq. (11). A *transversely* polarised target requires structure functions this model does not carry, so only the longitudinal case is written. And the hadronic tensor from which the first line is usually derived — eq. (10) of the radiative-correction note — is summed over the target spin and is built only from the photon basis vectors X, Y, ℓ ; the last two lines require the terms linear in the target spin vector S^μ , which that tensor does not have. The generator inherits the same restriction: it samples the first line only.

2 The data

Fourteen data sets are loaded from a single workbook, `pi0_eta_database.xlsx`, with two exceptions created in October 2026 and described below. Table 1 lists everything, fitted or not.

Table 1: The database and the fit. A set marked — is not fitted; its χ^2 is then a blind prediction. CLAS12 cross sections are preliminary and are the only blind set left.

set	key	fitted	points	χ^2	per point
CLAS6 π^0 p	clas6_pi0	yes	288	487.9	1.69
CLAS6 η p	clas6_eta	yes	207	202.2	0.98
Hall A 2017, π^0 n	halla_n	yes	8	16.4	2.05
Hall A 2017, σ_U at two ϵ	halla_n_U	yes	8	5.3	0.66
Hall A 2016, π^0 p	halla_y16	yes	21	102.4	4.88
Hall A 2016, σ_L separated	halla_y16_L	yes	7	4.5	0.64
Hall A 2011, π^0 p	halla_y11	yes	112	134.0	1.20
Hall A 2021, π^0 p	halla_y21	yes	144	346.6	2.41
CLAS12 cross sections	clas12_xs	—	438	5559.5	12.69
COMPASS 2025	compass	yes	26	65.9	2.53
Zhao, η BSA	bsa_zhao	yes	12	12.3	1.02
CLAS12 $\sigma_{LT'}/\sigma_0$	bsa_clas12	yes	30	48.1	1.60
De Masi, moment from $A_{LU}(\phi)$	bsa_demasi_mom	yes	60	61.8	1.03
eg1-dvcs, pol. target	eg1	yes	40	82.3	2.06

2.1 Two measurements recovered from the publications

For two days the author of this report stated that no separated σ_L existed anywhere in the database. That was true of the workbook and false of the literature, and the data were in the same two papers the fit was already using.

Proton σ_L , PRL 117 262001. The 2016 Rosenbluth separation published four structure functions; the workbook's `All_experiment.xlsx` has nine cross-section columns and none of them is σ_L , so seven points were silently absent from every fit up to and including September. They are the only separated σ_L that exists for this reaction. Three of the seven central values are negative, with errors of 400–460 nb/GeV² against values of the same size; the paper warns that σ_T and σ_L are strongly anticorrelated and the covariance is not published, so that warning cannot be honoured here. Their role in the fit is therefore one-sided: they do not measure σ_L , they *forbid a large one*.

Neutron σ_U at two beam energies, PRL 118 222002. E08-025 measured $\sigma_U = \sigma_T + \epsilon\sigma_L$ at $E = 4.455$ and 5.549 GeV ($\epsilon = 0.651$ and 0.791 , recomputed here and agreeing with the published values to three digits). The σ_T that the earlier fits used is *derived* from those two by the Rosenbluth solve and is constructed to have the longitudinal part removed, so it cannot see σ_L at all. The eight σ_U points therefore replace the four σ_T points; fitting both would be fitting one measurement twice. σ_{LT} and σ_{TT} keep their published values.

Both blocks are held in `halla_rosenbluth.py` rather than in the workbook, because the workbook's schema cannot express them.

2.2 De Masi: 60 moments, not 663 ϕ points

The De Masi beam-spin asymmetry (PRC 77 042201(R)) is published as *plots*; there is no table. The measurement is the ϕ distribution in the CLAS physics database — 703 points in 60 kinematic bins — and figure 5 is a rendering of it.

Until October 2026 the fit used those 703 points directly. That was wrong in one specific way: it supplied 663 points, 42 % of the whole fit, while carrying about *sixty* independent numbers. Measured rather than argued: scaling σ_{LT} in the denominator of the asymmetry by 0, 2 or -1 moves χ^2 by 0.4 to 3.1 units out of 625, and setting σ_{TT} to zero *improves* it by 12. The ϕ shape beyond $\sin \phi$ is not measured, and every χ^2/ndf quoted in the September report was flattered by about 0.3 as a result.

The moment is now extracted once, from all 703 points with $1/\sigma_{\text{stat}}^2$ weights, using the *full* asymmetry form

$$A_{LU}(\phi) = \frac{\sqrt{2\epsilon(1-\epsilon)} (\sigma_{LT'}/\sigma_0) \sin \phi}{1 + \sqrt{2\epsilon(1+\epsilon)} (\sigma_{LT'}/\sigma_0) \cos \phi + \epsilon (\sigma_{TT}/\sigma_0) \cos 2\phi} \quad (13)$$

rather than a plain $A \sin \phi$ fit. In 2008 the denominator was unknown, so a plain fit was the only option; today σ_U and σ_{TT} are measured at these very kinematics (CLAS6 π^0 , mean pull -0.10 and $+0.09$), so the extracted number is the physical $\sqrt{2\epsilon(1-\epsilon)} \sigma_{LT'}/\sigma_0$ and not its projection onto $\sin \phi$. The two differ by 7.6 %.

The extraction is model-independent in practice. Swapping the amp2609 denominator for this fit's moves the extracted amplitude by a median 0.009 of its statistical error, at most 0.044. The denominator needs only σ_{LT}/σ_0 and σ_{TT}/σ_0 , which the cross sections pin and on which both fits agree.

Errors are statistical only. The published 0.016 names two sources, “event selection plus the choice of the fit function”. The second is removed by construction here — there is no longer a choice — and its size can be measured as $|A_{\text{full}} - A_{\text{sin}}|$, median 0.0074, about half the quoted systematic. A ϕ -independent first source barely reaches the moment: the leakage $\sum w \sin \phi / \sum w \sin^2 \phi$ has median 0.023, so 0.016 moves the amplitude by 0.0004 against statistical errors of 0.02–0.08. The 3.5 % beam polarisation remains a fitted nuisance, which is where a fully correlated scale belongs.

The digitised moments are gone from the database. The workbook's own `datasets` sheet records that its 62 moment values came off figure 5 and that their errors are 1.5 times too small.

They also carry bin labels instead of measured means — $Q^2 = 1.30$ where the mean is 1.14 — and nine of the sixty measured bins have no partner among them. The extraction above reproduces them to a median 2.0 %, which is the accuracy of reading a plot, so it is closer to what was published than the digitisation of what was published; they were dropped from the database on 5 October 2026.

2.3 Provenance of the uncertainties

Every weight in the fit comes from a publication. Three defects repaired in September are recorded here because they changed the answer.

The Hall A neutron systematics exist only as a picture. PRL **118** 222002 gives no table; the point-to-point uncertainties appear only as the shaded bands of its figure 5. They were read from the vector PDF, with the vertical calibration taken not from the axis but from the published error bars themselves. The calibration closes to 0.0 % in the σ_T panel and 1.8 % in σ_{LT} and σ_{TT} . The bands are asymmetric, in places by a factor three; they are symmetrised as $(\text{up} + \text{down})/2$, and the 3.1 % normalisation they contain is removed in quadrature because it is fitted separately.

An invented 10 % was removed. Hall A 2016 and the neutron had been carrying a flat 10 % of the author’s own making. It was wrong in both directions at once: it concealed a real discrepancy in the 2016 proton set and inflated a false one in the neutron, whose true bands are larger than 10 %. An invented uncertainty is worse than a missing one, because it looks like knowledge.

2.4 Normalisation uncertainties

Three sets carry an overall normalisation that multiplies every point together: 3.5 % for De Masi, 3.12 % for Hall A 2016 (PRL **117** 262001, table III) and 3.1 % for the neutron. Adding these in quadrature to the per-point errors would discard the correlation and smear each set instead of letting it move as a whole. They are fitted as one free scale per set,

$$r = \frac{\text{data} - \lambda \text{ model}}{\sigma}, \quad \text{with a penalty} \quad \left(\frac{\lambda - 1}{\sigma_{\text{norm}}} \right)^2 \text{ counted in } \chi^2, \quad (14)$$

and the number of degrees of freedom reduced by the number of λ .

For an asymmetry this deserves a word, since luminosity and acceptance cancel in a ratio. What does not cancel is the beam polarisation: A_{LU} is divided by a measured P_{beam} , and its 3.5 % scales every point of the set together. The λ on De Masi is therefore a polarimetry scale, not a cross-section normalisation. It is also nearly degenerate with the overall size of $\sigma_{LT'}$, far more so than for a cross section, so its excursion should be read as a diagnostic rather than as a calibration: see section 7.

Table 2: Fitted normalisations.

set	prior	λ	pull
bsa_demasi_mom	3.50 %	1.0381 ± 0.0322	$+1.09 \sigma$
halla_n	3.10 %	0.9876 ± 0.0305	-0.40σ
halla_y16	3.12 %	1.0519 ± 0.0260	$+1.66 \sigma$

3 The fit

3.1 Constraints

Three constraints are imposed for stated physical reasons rather than to improve χ^2 .

$n_{Q,d} = n_{Q,u}$ **in both transverse sectors.** A generalised form factor is (GPD) $\times$ (hard subprocess). The GPD carries no Q^2 dependence, and the subprocess is flavour-blind up to the quark charge. Transversity does not mix with gluons, so its evolution is non-singlet and flavour-blind as well. Left free, the fitted $n_{Q,d} - n_{Q,u}$ came out -7.0 and $+2.8$ in successive fits, with flavour ratios running by an order of magnitude across Q^2 .

$b_d = b_u$ **for H_T .** Left free, the fit drives b_d to whatever ceiling it is given — 12, 18, 25 — buying two units of χ^2 while the normalisation N_d compensates by a factor sixteen. It is a flat valley, not a measurement.

$\langle \tilde{E} \rangle$ **follows $\langle \tilde{H} \rangle$ in shape.** Both b' and n_Q are tied; the reason is in paragraph 1.2 and is a statement about the region beyond the data, not about χ^2 .

In addition every t -slope is required non-negative over $x_B \in [0.05, 1]$, not only where there are data: without it H_T^u came out with $b = -0.26$, so that the form factor *grew* with $|t|$ above $x_B = 0.785$.

3.2 Parameters

Tables 3 and 4 give all 30 fitted parameters with errors from the Jacobian at the minimum. A dagger marks a parameter sitting on a bound, where the quoted error is one-sided and the value is not a measurement.

Table 3: Block parameters. Every block is a *convolution* $\langle X \rangle$, not a GPD: no GPD is fitted anywhere in this work. N in $\text{nb}^{1/2}/\text{GeV}$, b and b' in GeV^{-2} , n_Q dimensionless.

block	N	b	b'	n_Q
$\langle H_T \rangle^u$	20.397 ± 0.637	0.001 ± 0.057	-1.305 ± 0.090	1.297 ± 0.040
$\langle H_T \rangle^d$	-1.962 ± 1.770	$= b_u$	$= b'_u$	$= n_{Q,u}$
$\langle \bar{E}_T \rangle^u$	72.177 ± 3.231	0.743 ± 0.052	-0.368 ± 0.036	1.123 ± 0.060
$\langle \bar{E}_T \rangle^d$	200.442 ± 21.866	5.155 ± 0.385	$= b'_u(\bar{E}_T)$	$= n_{Q,u}(\bar{E}_T)$
$\langle \tilde{E} \rangle$	30.018 ± 3.772	$0.000 \pm 0.078^\dagger$	$= b'(\tilde{H})$	$= n_Q(\tilde{H})$
$\langle \tilde{H} \rangle$	8.725 ± 1.417	0.000 ± 0.085	-0.664 ± 0.067	0.732 ± 0.152

Table 4: Remaining parameters, with the interval each was allowed.

parameter	value	lower	upper
$R_L = N_d/N_u (\langle L \rangle)$	-0.802 ± 0.315	-2.00	2.00
δ_0	1.307 ± 0.115	-3.14	3.14
δ_1	0.452 ± 0.091	-3.00	3.00
ρ_{CE}	0.245 ± 0.023	0.00	1.50
ϕ_{CE} (frozen)	0.000 ± 0.000	-3.14	3.14
ϕ_w	-0.245 ± 0.080	-3.14	3.14
ρ_{nf}	$0.000 \pm 0.000^\dagger$	0.00	3.00
ϕ_{nf}	2.729 ± 0.129	-3.14	3.14
$\varphi_{d/u}(\langle H_T \rangle)$	-0.429 ± 2.516	-3.14	3.14
$\varphi_{d/u}(\langle \bar{E}_T \rangle)$	0.454 ± 0.264	-3.14	3.14
$\varphi_{d/u}(\langle L \rangle)$	0.954 ± 0.206	-3.14	3.14

3.3 Derived quantities

At $Q^2 = 1.75 \text{ GeV}^2$, $x_B = 0.36$, $-t = 0.27 \text{ GeV}^2$:

$$\left. \frac{d}{u} \right|_{H_T} = -0.096, \quad \left. \frac{d}{u} \right|_{\bar{E}_T} = 0.739, \quad \left. \frac{n}{p} \right|_{\sigma_{TT}} = 0.889, \quad \frac{\sigma_L}{\sigma_T} = 0.132.$$

The n/p ratio carries a caveat that is not a formality; see section 7.

4 The longitudinal sector

4.1 What $\langle \tilde{H} \rangle$ buys

With the block in place, at $Q^2 = 2.5 \text{ GeV}^2$ and $x_B = 0.2$,

$$\frac{\sigma_L}{\sigma_T} = 0.093 \text{ at } t' = 0.002 \text{ GeV}^2, \quad 0.106 \text{ at } t' = 0.3 \text{ GeV}^2.$$

Before it, the first number was zero by construction. That is the whole physical case for the block, and it does not depend on χ^2 : a longitudinal non-flip amplitude with $\Delta = 0$ cannot vanish at $t' = 0$.

4.2 What the data say about its size

The χ^2 case is weaker and took three attempts to measure honestly, because this surface has many local minima and a single fit measures effort rather than physics. Continuation chains were run on both sides, each node seeded from the previous: ten nodes for the 37-parameter model without $\langle \tilde{H} \rangle$ and twelve with it.

	free parameters	χ^2 floor	spread over nodes
without $\langle \tilde{H} \rangle$ (Mom_A9)	28	1596.0	22.8
with $\langle \tilde{H} \rangle$ (Mom_B12)	30	1573.8	8.0 (last six)

$\Delta\chi^2 = 22.1$ for two extra parameters, in favour of the block. The spread is one-sided — nodes that failed to return to the floor — and each model reached its own floor more than once, so the floors are reliable even though the spread is as large as the gap. The block wins about 14 units each on CLAS6 η , Hall A 2011 and Hall A 2021, and loses 12 on CLAS6 π^0 and 9 on eg1: a redistribution, not a uniform gain.

A retraction. An earlier version of this work stated that the recovered Rosenbluth data make the $\langle \tilde{H} \rangle$ model 9.5 units *worse*. That number came from single fits run to no particular convergence. On the present input a single such fit gives 1721 where the chain floor is 1574: 147 units of pure effort. The claim is withdrawn. It measured minimisation, not physics, and it is recorded here because the same trap will be there next time.

5 σ_{LT} , and where its sign comes from

This section is a joint result with an independent fit of the Goloskokov–Kroll model (`libGKPi0`), each side having run the other’s test.

5.1 Two terms, not one

By eq. (10), $\sigma_{LT} = -\sqrt{2} \text{Re} [T_{00}^{(+)*} T_{01}^{(+)} + T_{00}^{(-)*} T_{01}^{(-)}]$: a non-flip term and a flip term. In GK the line `result.Mppm0 = result.Mppp0` makes the non-flip interference vanish *identically* — that is

the $C = E$ relation — so GK’s σ_{LT} is one term where this model has two. Here $C = E$ is unlocked by $\rho_{CE} = |T_{01}^{(+)}|/|U_{01}^{(+)}|$, fitted.

The two terms largely cancel. For π^0 off the proton at $x_B = 0.36$:

$Q^2, t $ [GeV ²]	flip term	non-flip term	σ_{LT}
1.50, 0.228	−87.4	+73.8	−13.6
1.75, 0.232	−56.6	+45.3	−11.4
2.00, 0.235	−38.7	+29.7	−9.1

This is why σ_{LT} is the observable everyone gets wrong: it is the small difference of two numbers several times larger.

$C = E$ is load-bearing, and only σ_{LT} sees it. Setting $\rho_{CE} = 0$ at $Q^2 = 1.75$, $|t| = 0.232$ moves σ_T by 1.1 %, σ_{TT} by 5 %, σ_L not at all — and σ_{LT} by a factor three. GK cannot test $C \neq E$ even in principle, since no parameter is attached to that line, so ρ_{CE} is the only number either framework has on it, fitted to seven points of which two are the pair discussed below.

5.2 The observable angle, and where this model sits

σ_{LT} and $\sigma_{LT'}$ are two coordinates of a point with a two-dimensional error. The quantity free of every convention is the angle of the line they lie on,

$$\delta_{\text{obs}} = \text{atan2}(\sigma_{LT'}, \sigma_{LT}) \bmod 180^\circ. \quad (15)$$

It must be estimated by profiling the likelihood in that angle, not by averaging the ratio $\sigma_{LT'}/\sigma_{LT}$: there $\sigma_r \propto |r|$ makes the weight go as $1/r^2$, and dividing by a σ_{LT} with large errors is regression dilution. Four estimators on the same data span -0.58 to -1.64 . The profile

$$\chi^2(\delta) = \sum_i \frac{(y_i \cos \delta - x_i \sin \delta)^2}{(\sigma_{y,i} \cos \delta)^2 + (\sigma_{x,i} \sin \delta)^2}, \quad x = \sigma_{LT}, \ y = \sigma_{LT'}, \quad (16)$$

divides nothing, needs no cut, uses every point, and stays bounded where the ratio runs away.

	δ_{obs}	68 %	χ^2/ndf
data, Hall A 2011	121.4°	[116.3, 127.2]	0.78
data, Hall A 2021	126.6°	[123.1, 130.1]	2.03
this fit , at the same points	101.8° / 115.8°	rms 3.7 / 11.3	
GK, from its delivered structure functions	57°		

The data place the ray firmly in the second quadrant — $\sigma_{LT} < 0$, $\sigma_{LT'} > 0$: at 2011, 13 of 14 significant σ_{LT} are negative and 16 of 16 $\sigma_{LT'}$ positive; at 2021, 25 of 28 and 26 of 27. GK sits in the *first* quadrant, both positive.

So the ranking is data 121–127°, this model 102–116°, GK 57°. **This model is 10–20° out and outside the intervals**; GK is 65–75° out and in the wrong quadrant, and its $\sigma_{LT'}$ is additionally a factor 4.4 small — one statement, not two: its $\langle \tilde{E} \rangle^* \langle H_T \rangle$ product is too small and rotated by about 70°.

A comparison trap, recorded. An earlier version of this section quoted GK’s *internal* phase $\arg \langle \tilde{E} \rangle - \arg \langle H_T \rangle = 122\text{--}124^\circ$ and compared it with a $\pm 90^\circ$ window. That is not observable: the prefactors on the real and imaginary parts carry different signs, so the delivered structure functions sit at $180^\circ - \delta$. Quoted as a property of GK, 122° would have said GK agrees with the data. It does not. Only eq. (15), built from the two structure functions as a code delivers them, can be compared between frameworks.

5.3 Three points that no smooth model reaches

Where the χ^2 sits, over the seven Hall A 2016 σ_{LT} points:

	$Q^2 = 1.50$ pair	(1.75, 0.281)	the other four
this fit (40.3 total)	27.1 = 67 %	8.2 = 20 %	5.0 = 13 %
GK, sign flipped by hand (33.0)	24.1 = 73 %	6.5 = 20 %	2.3 = 7 %

The same three points, in the same order, from models built on different assumptions with flip amplitudes a factor ten apart in relative size. Four of seven are described by both; three by neither.

Look at the data alone, at the same x_B and nearly the same $|t|$: $\sigma_{LT} = -100.1$ and -169.2 at $Q^2 = 1.50$; -16.9 , -17.2 and $+13.1$ at 1.75; -1.6 and -6.2 at 2.00. A factor six between the first two Q^2 and another three to the next is an effective power near Q^{-13} , which no twist expansion produces; and the point at (1.75, 0.281) is positive where its two neighbours at the same Q^2 are negative.

The claim worth making is the weak one: *no smooth model reaches these three points*. Not that the measurements are wrong.

6 What the leave-one-out analysis shows

Each fitted set was removed in turn and the fit repeated from the reference minimum. A set's χ^2 when it is *not* fitted is a blind prediction, and is the useful number: a set that is left out and still described brings no unique information.

Table 5: Blind χ^2 per point of the removed set, against its χ^2 per point in the reference fit, and the effect on two derived quantities. Sorted by blind χ^2 .

removed	points	blind χ^2 /pt	in the fit	n/p	σ_L/σ_T
reference (nothing removed)	—	—	—	0.889	0.132
Hall A 2016 $\sigma_T, \sigma_{LT}, \sigma_{TT}$	21	6.04	4.88	0.880	0.154
eg1-dvcs	40	6.03	2.06	0.870	0.188
Hall A 2021	144	3.33	2.41	0.837	0.119
CLAS12 BSA	30	2.94	1.60	0.890	0.133
COMPASS	26	2.89	2.53	0.617	0.169
CLAS6 π^0	288	2.50	1.69	0.806	0.108
Hall A neutron σ_{LT}, σ_{TT}	8	2.38	2.05	0.928	0.131
De Masi moment	60	1.68	1.03	0.889	0.121
Zhao η BSA	12	1.55	1.02	0.909	0.142
Hall A 2011	112	1.34	1.20	0.907	0.128
CLAS6 η	207	1.03	0.98	0.923	0.140
Hall A neutron σ_U	8	0.69	0.66	0.893	0.133
Hall A 2016 σ_L	7	0.64	0.64	0.883	0.135

Nothing is irreplaceable any more, and that is new. In the September version, dropping the CLAS6 η data left the model predicting η at 219 per point. It is now 1.03 — the same as its 0.98 inside the fit. Hall A 2021 reaches $x_B = 0.60$ and has taken over the region that only the η cancellation used to probe. The largest blind χ^2 in the table is 6.0, for Hall A 2016 and for eg1.

eg1 still fixes a phase, but less violently. Its blind 6.03 against 2.06 in the fit is the largest relative jump of any set, which is the $\sigma_{T'}$ interference of section 7: no other observable in the database sees it. The September figure was 40 per point; the difference is that $\langle \tilde{H} \rangle$ now carries part of the longitudinal sector that ϕ_w used to absorb.

COMPASS moves n/p more than the neutron does. Removing the eight neutron σ_{LT}, σ_{TT} points moves n/p from 0.889 to 0.928; removing COMPASS moves it to 0.617 and $d/u|_{H_T}$ from -0.096 to -0.302 , by far the largest excursion in the table. Twenty-six COMPASS points at $x_B \approx 0.09$, an order of magnitude lower than anything else, do more to the flavour separation than the only neutron data that exist. That is worth stating plainly: the n/p ratio is not measured by the neutron.

The two recovered sets behave as one-sided constraints. Both σ_L (blind 0.64 against 0.64 fitted) and the neutron σ_U (0.69 against 0.66) are predicted as well when left out as when fitted, and removing either moves nothing. That is the expected signature of points whose errors are the size of their values: they forbid, they do not measure.

7 What the data determine weakly, or not at all

n/p is soft against the model, not only against the data. The neutron channel is $\langle X \rangle \propto X_u + 2X_d$ against the proton's $2X_u + X_d$, so the d block enters the neutron with twice the weight it has in the proton. Since $\sigma_{TT} = \sum |T_{01}|^2 - \sum |U_{01}|^2$ and U_{01} is built from $\bar{E}_T$, the ratio n/p in σ_{TT} is essentially a statement about $\bar{E}_T^d$ relative to $\bar{E}_T^u$.

This fit gives 1.16, 1.00 and 0.86 at the three $|t|$ where a proton point exists, against a measurement whose point-by-point values are 0.41 ± 0.52 , 0.33 ± 0.23 , 0.24 ± 0.20 . Two remarks keep that from being a verdict. First, the published single number 0.28 ± 0.07 assumes *zero* u - d relative phase; PRL **118** 222002 says so explicitly and carries the phase as a systematic that makes its figure 6 bands enormous. At a phase near π the same 0.28 is reached with $|\bar{E}_T^d/\bar{E}_T^u| = 0.81$, which is where both this model and GK already sit. Second, the ratio is *not* flat in $|t|$: GK gives 0.790/0.797/0.798, while every fit that has seen the neutron data falls by a factor 1.4, as the data themselves do by 1.67. A ratio quoted as one number hides that.

What costs what: a penalty dragging the ratio down from 0.99 to 0.67 costs 85 units of χ^2 , to 0.43 costs 130, to 0.34 costs 181. The neutron sets gain only 7 units; the proton sets and CLAS6 π^0 lose 190. In this fit the ratio is held *up* by the proton, not by the neutron.

One experiment fixes a phase. $\sigma_{T'} = 2 \text{Re}\langle T_{01}, U_{01} \rangle$ by eq. (10); the unpolarised data fix only the moduli. The A_{LL} moments of eg1-dvcs are the only observable in the database that sees this interference, and when eg1 is left out its own blind χ^2 is 41 per point, with ϕ_w swinging from -0.70 to $+2.90$ — essentially a sign flip of $\sigma_{T'}$. Forty points from one experiment.

The experiments do not agree on the sign of σ_{LT} . The blind prediction of the preliminary CLAS12 cross sections gives 32 per point in σ_{LT} against 2.8 in σ_U and 3.2 in σ_{TT} . Looking at the data rather than at the χ^2 : all three Hall A measurements are firmly negative (mean σ_{LT} of -77.6 , -42.6 , -7.0 , with 0.83–0.86 of points negative), CLAS6 sits near zero, and CLAS12 is positive ($+8.3$, only 0.24 negative). The model follows the sets that are in the fit. The pull grows monotonically with x_B — a pattern that looks more like a convention than like physics — and the CLAS12 numbers are preliminary.

$\sigma_{LT'}$ is too small, in three independent places. The De Masi moments sit above the model, and the normalisation nuisance is dragged to $\lambda =$ trying to follow them. That is not a calibration: where $\sigma_{LT'}$ is measured *directly* rather than in a ratio, Hall A 2011 and Hall A 2021 give mean pulls $+0.25$ and $+0.27$, in the same direction. Meanwhile σ_0 is unbiased — CLAS6 π^0 σ_U has mean pull -0.10 — so the deficit is in the numerator, not the denominator. Three independent indications that the model's $\sigma_{LT'}$ is about 30% too small.

A single power of Q^2 does not span the range. Hall A 2016 and Hall A 2021 disagree with the model in opposite directions at the two ends. Fitting a power to the Hall A 2016 separated σ_T at $|t| = 0.23$ gives $\sigma_T \sim Q^{-6.3 \pm 1.2}$, against the twist-3 expectation Q^{-8} ; this model gives $Q^{-5.5}$ over $Q^2 = 1.5\text{--}6$, while its σ_L comes out $Q^{-6.5}$ against the twist-2 Q^{-6} . The longitudinal power is close to the counting; it is σ_T that is far too flat, in the data as well as in the model, and that is what keeps $R = \sigma_L/\sigma_T$ from growing as Q^2 in the measured range.

8 The generator

The model is installed in `exclurad.py` as `amp2613` across all four channels, π^0 and η off the proton and off the neutron. Three checks are worth recording because each caught something.

Closure. `exclurad.py` keeps three private copies of the amplitude code. Dropping a 43-slot parameter file into it raises no error and gives neither the old model nor the new one — σ_L out by a factor five, because slot 25 is now dead and the refitted $\langle \tilde{E} \rangle$ is calibrated to work alongside $\langle \tilde{H} \rangle$. The closure test compares all three copies against this repository’s `amplitudes.py` over 598 points in all four channels; they agree to 3.7×10^{-14} .

The radiative integral. $\eta = \sigma_{\text{obs}}/\sigma_{\text{Born}}$ at the node $W^2 = 16$, $Q^2 = 1.15$, $t = -0.3$, $h = +1$ is 14.4 for the model without $\langle \tilde{H} \rangle$, 577 for the first tied variant and 194 for `amp2613`. The first *untied* variant gave 2.6×10^9 and ran eleven times slower; that is what the shape tie of paragraph 1.2 repairs.

The stage-2 bound. Built per box with stratified probing — every cell probed equally, rather than from the proposal, which never reaches the tail cells — the bound is 2.64, 1.63, 8.34 and 2.03 for $\pi^0 p$, ηp , $\pi^0 n$, ηn . An overnight campaign of eight samples of 100 000 events, radiative and Born, unpolarised target, $v_{\text{cut}} = 0$, completed with *zero* bound violations in 160 jobs and 515 ms per event, against 464 ms for the model without $\langle \tilde{H} \rangle$ and 5256 ms for the untied variant.

9 Comparison with Goloskokov–Kroll

Beyond section 5, two comparisons put the GK prediction for $\sigma_{LT'}$ far below the data. The published GK curve for the η beam-spin asymmetry (Zhao *et al.*, PLB **789** 426, figure 5) is near zero across all four bins where the data are 0.1 to 0.3. For π^0 , a GK calculation over the 60 De Masi bins gives 0.01–0.025 against a measured median of 0.080. The present model gives a median 0.077 on the same bins, but that is a fit, not a prediction.

One asymmetry of the comparison should be stated. In GK the longitudinal sector has no free parameter at all: $\langle \tilde{H} \rangle$ and $\langle \tilde{E} \rangle$ are hard-coded, and the sixteen fitted parameters are H_T and $\tilde{E}_T$ only. The seven recovered proton σ_L points are therefore a prediction test there and a constraint here — an independent fit of GK gives $\chi^2 = 4.48$ on them for three different parameter sets, identical to the last digit.

The same is true of $\sigma_{LT'}$, and it was measured. Refitting all sixteen GK parameters against the 60 moments of section 2.2 together with the cross sections moves the moments from 6.24 to 5.84 per point, the total χ^2 by 27 units out of 1873, and no parameter by more than 0.23σ ; most of the gain came through the polarisation nuisance, which ends at $+3.4\sigma$ of its 3.5 % prior. With $M_{\text{pmp0}} = M_{\text{ppp0}}$, $\sigma_{LT'}$ is $|\langle \tilde{E} \rangle| |\langle H_T \rangle| \sin \delta$; $\langle \tilde{E} \rangle$ is frozen, $|\langle H_T \rangle|$ is held by σ_T , and δ is the frozen phase. No free parameter touches the product.

A success that was not one. GK fits asymmetries *alone* at 1.03 per point and cannot fit them together with the cross sections. Fitted alone it rescales $\langle H_T \rangle$ freely — an asymmetry is a ratio and

cannot see an overall scale — and lands on a negative κ_T for both flavours. Fitted together, σ_T holds $\langle H_T \rangle$ and the asymmetries take whatever $\langle \tilde{E} \rangle$ leaves them. The lesson is general and is the reason section 2.2 matters beyond bookkeeping: a set of ratios fitted on its own can be described by a normalisation the ratios do not constrain, and the 703 ϕ points hid at 1.34 per point what the 60 moments show at 6.24, on the same measurement.

10 Figures

Every figure shows the data of one set against the fit. A title reading NOT IN THE FIT marks a blind prediction. Data are drawn in red, the model in blue. Each panel carries its own kinematics and χ^2 ; the model curve includes the fitted normalisation of table 2 where the set has one, so the curve is what the fit actually compares against the points.

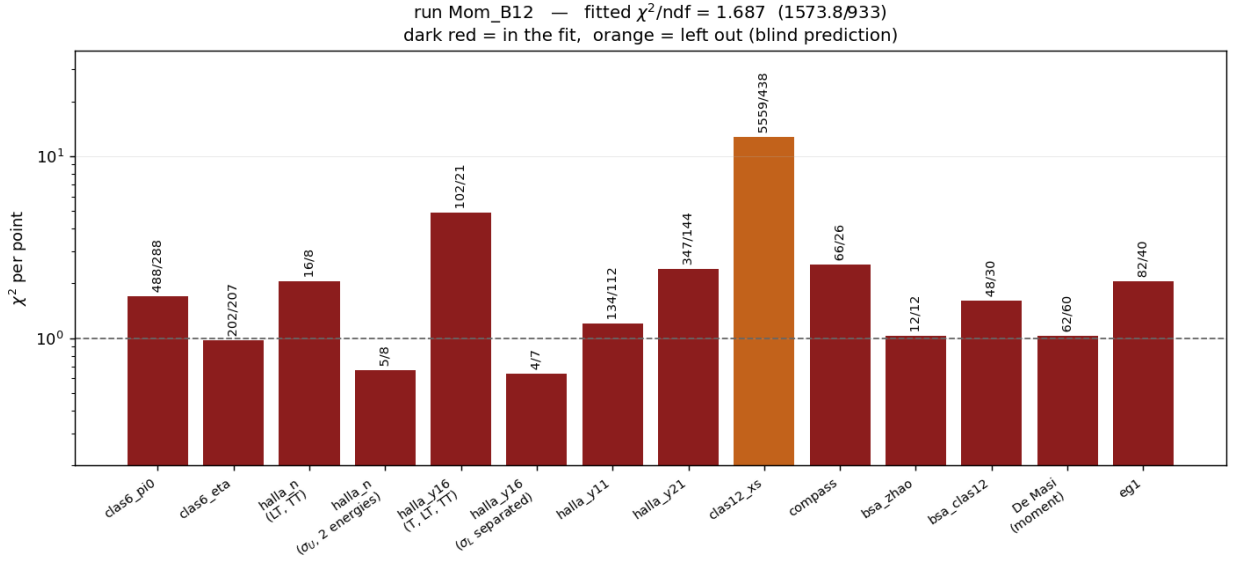


Figure 1: All fourteen sets at a glance: χ^2 per point, fitted and blind.

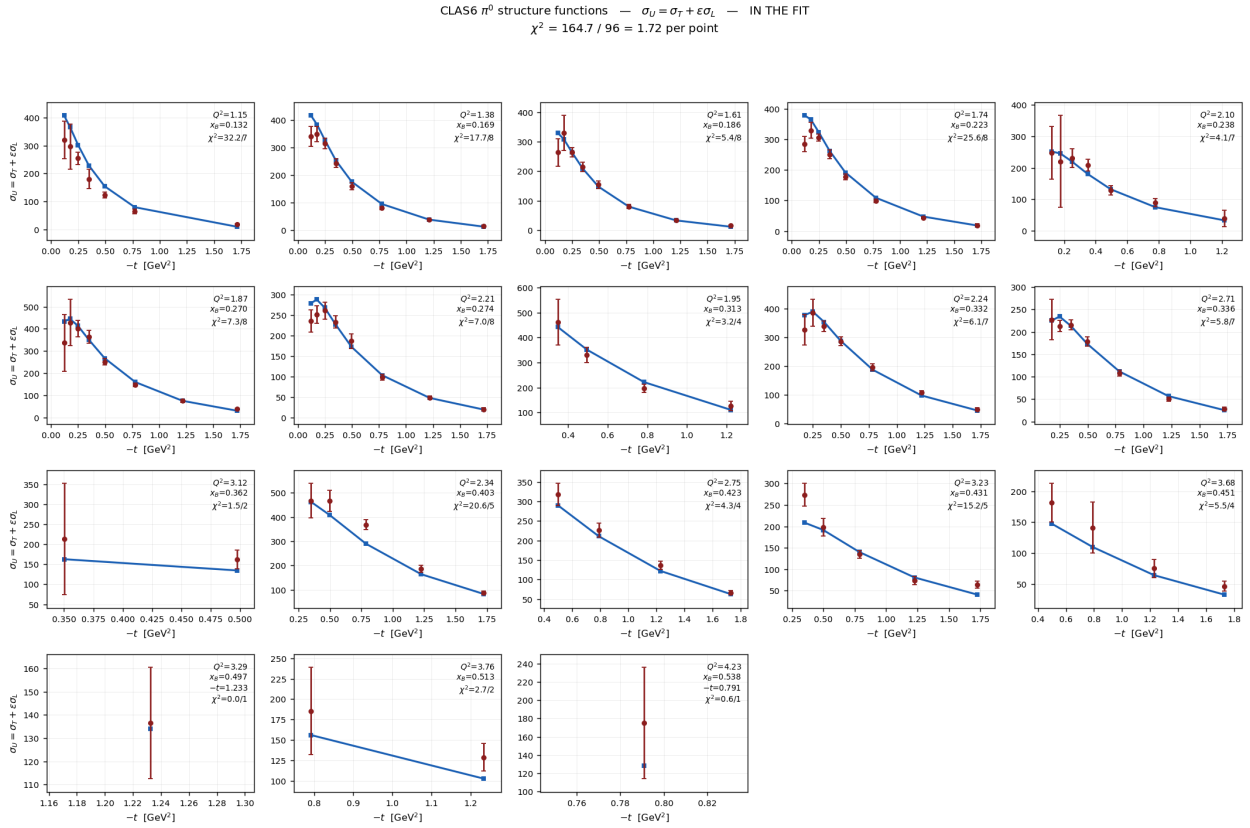


Figure 2: CLAS6 π^0 , $\sigma_U = \sigma_T + \epsilon\sigma_L$, with re-computed radiative corrections.

CLAS6 π^0 structure functions — σ_{LT} — IN THE FIT
 $\chi^2 = 207.7 / 96 = 2.16$ per point

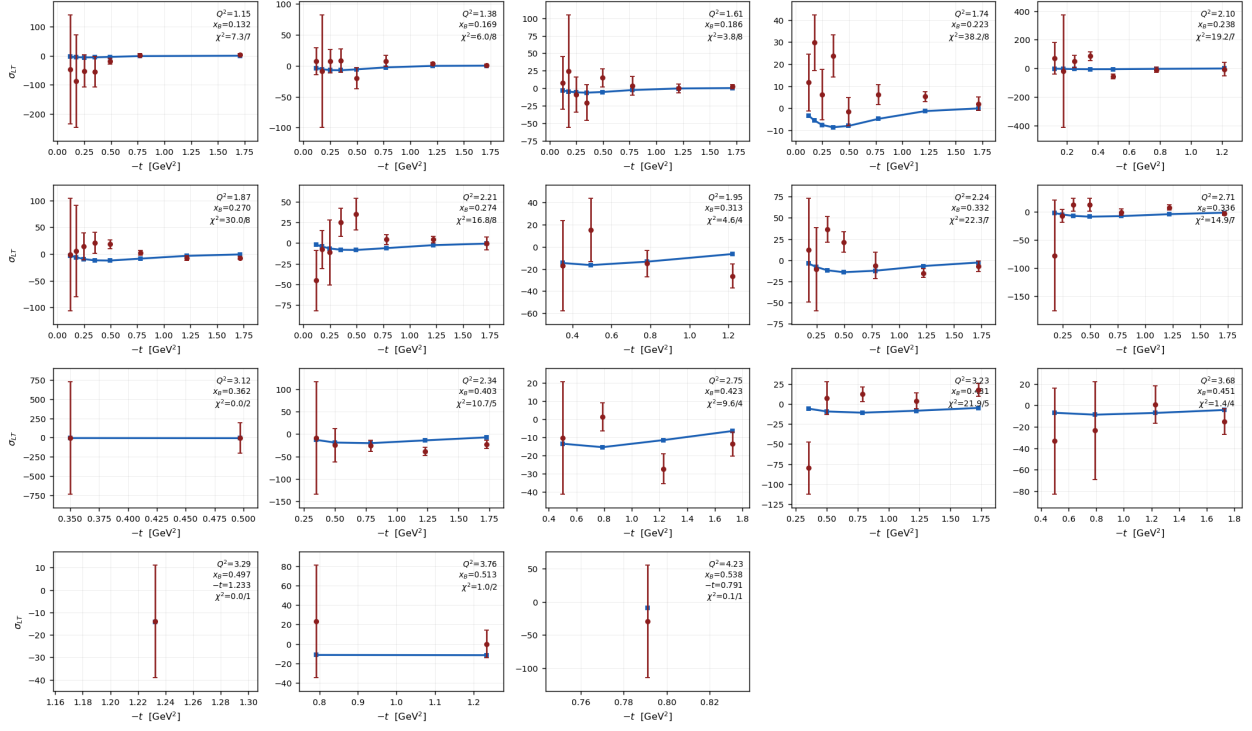


Figure 3: CLAS6 π^0 , σ_{LT} .

CLAS6 π^0 structure functions — σ_{TT} — IN THE FIT
 $\chi^2 = 115.5 / 96 = 1.20$ per point

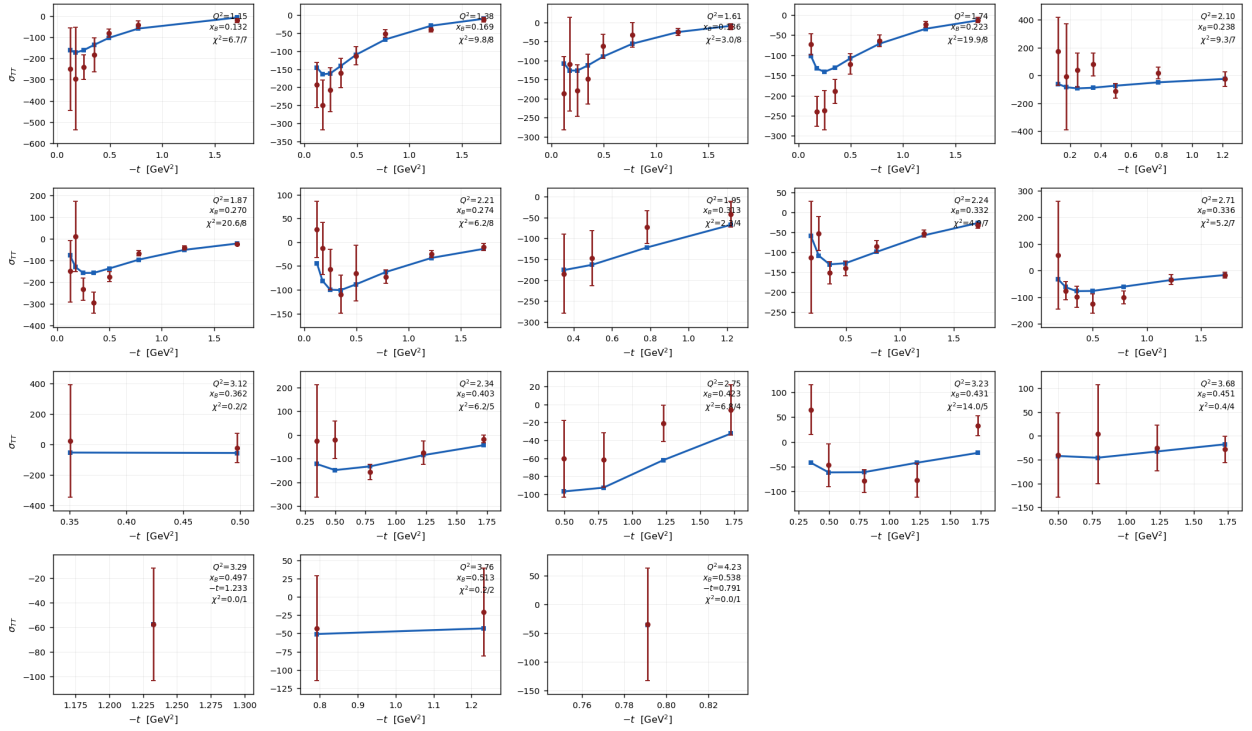


Figure 4: CLAS6 π^0 , σ_{TT} .

CLAS6 η structure functions — $\sigma_U = \sigma_T + \epsilon \sigma_L$ — IN THE FIT
 $\chi^2 = 91.3 / 69 = 1.32$ per point

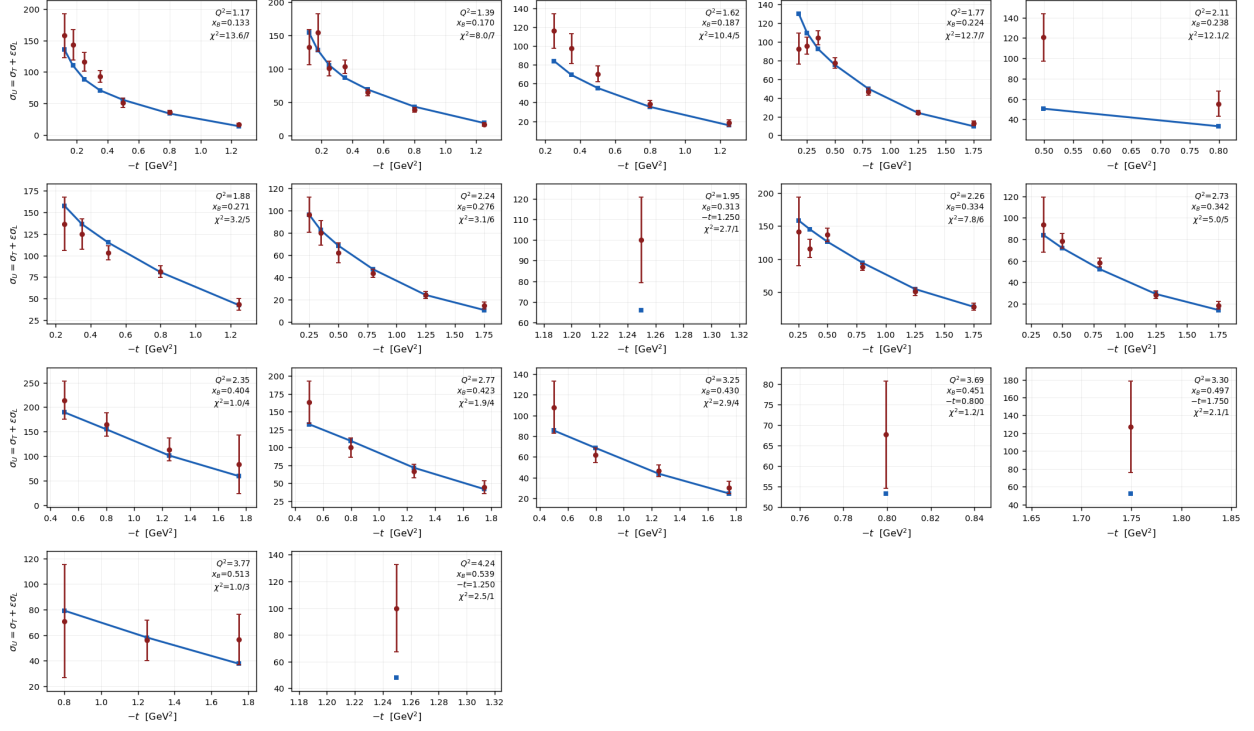


Figure 5: CLAS6 η , σ_U . This channel probes $2X_u - X_d$.

CLAS6 η structure functions — σ_{LT} — IN THE FIT
 $\chi^2 = 47.9 / 69 = 0.69$ per point

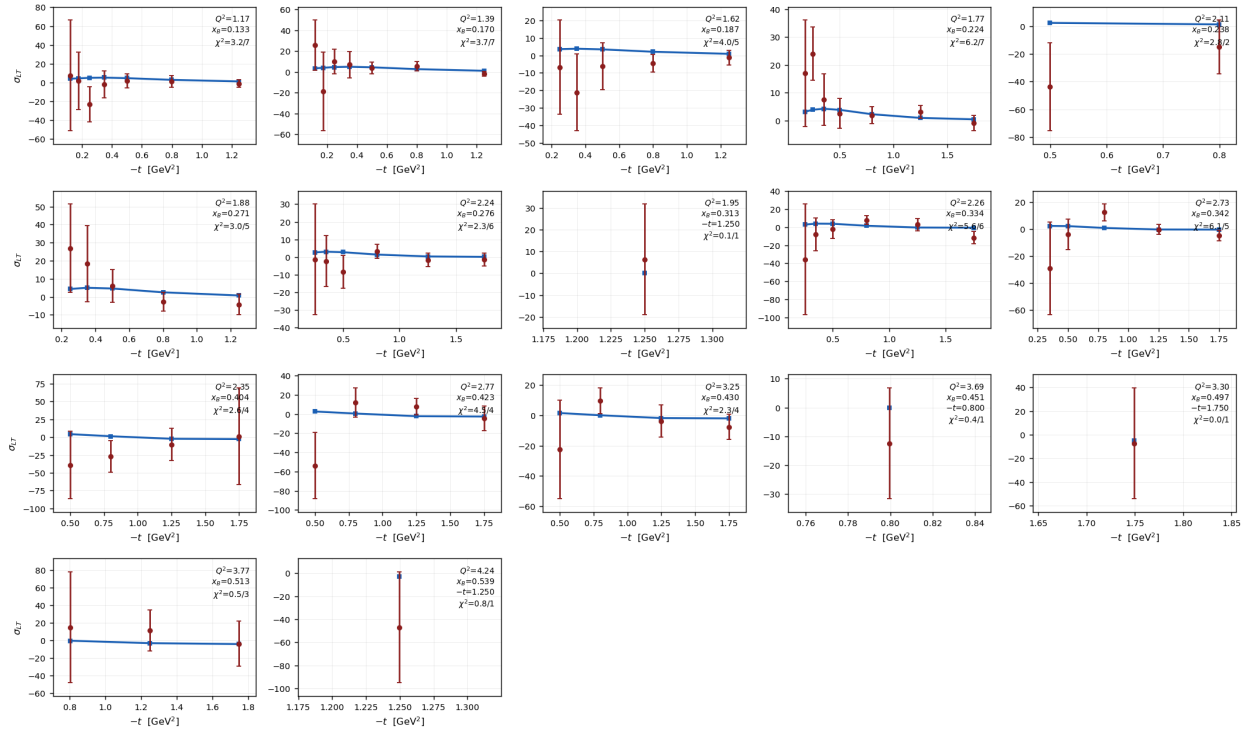


Figure 6: CLAS6 η , σ_{LT} .

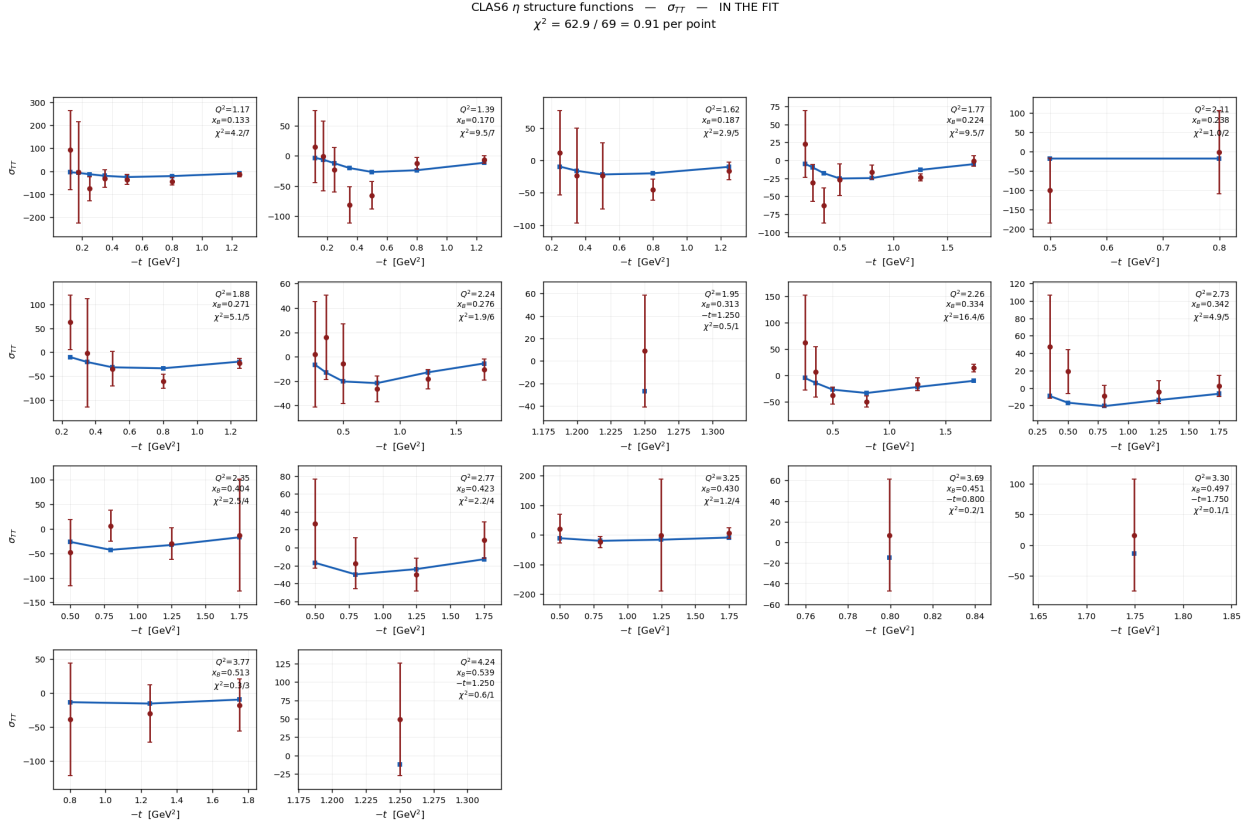


Figure 7: CLAS6 η , σ_{TT} .

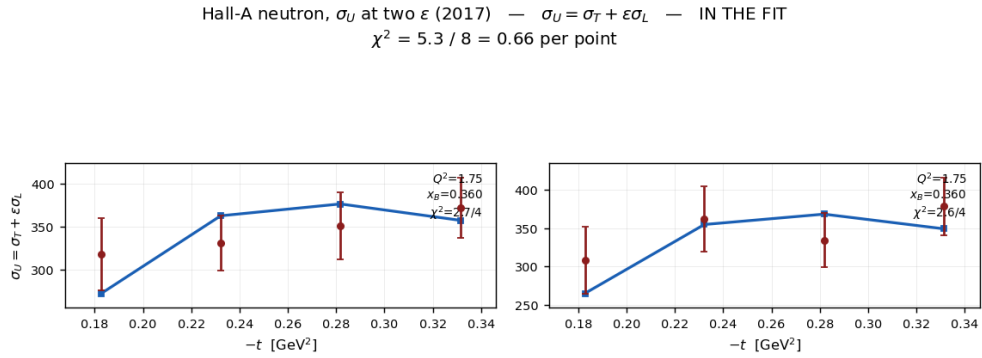


Figure 8: Hall A neutron, σ_U at the two beam energies. Recovered from PRL **118** 222002, section 2.1; these eight points replace the four derived σ_T .

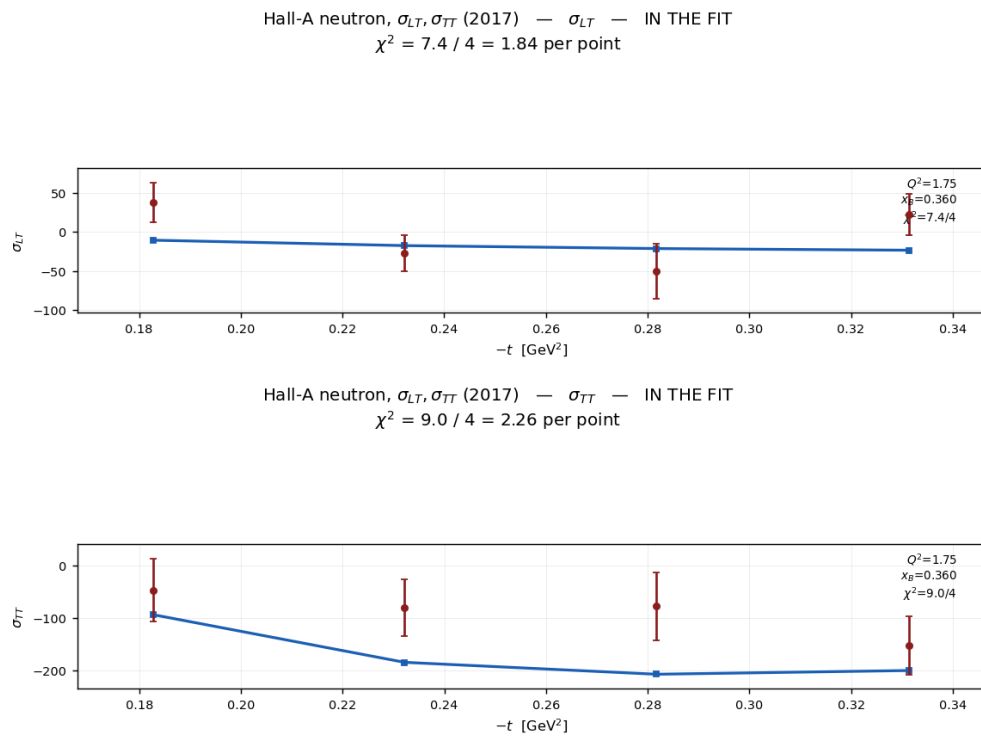


Figure 9: Hall A neutron, σ_{LT} and σ_{TT} , published values.

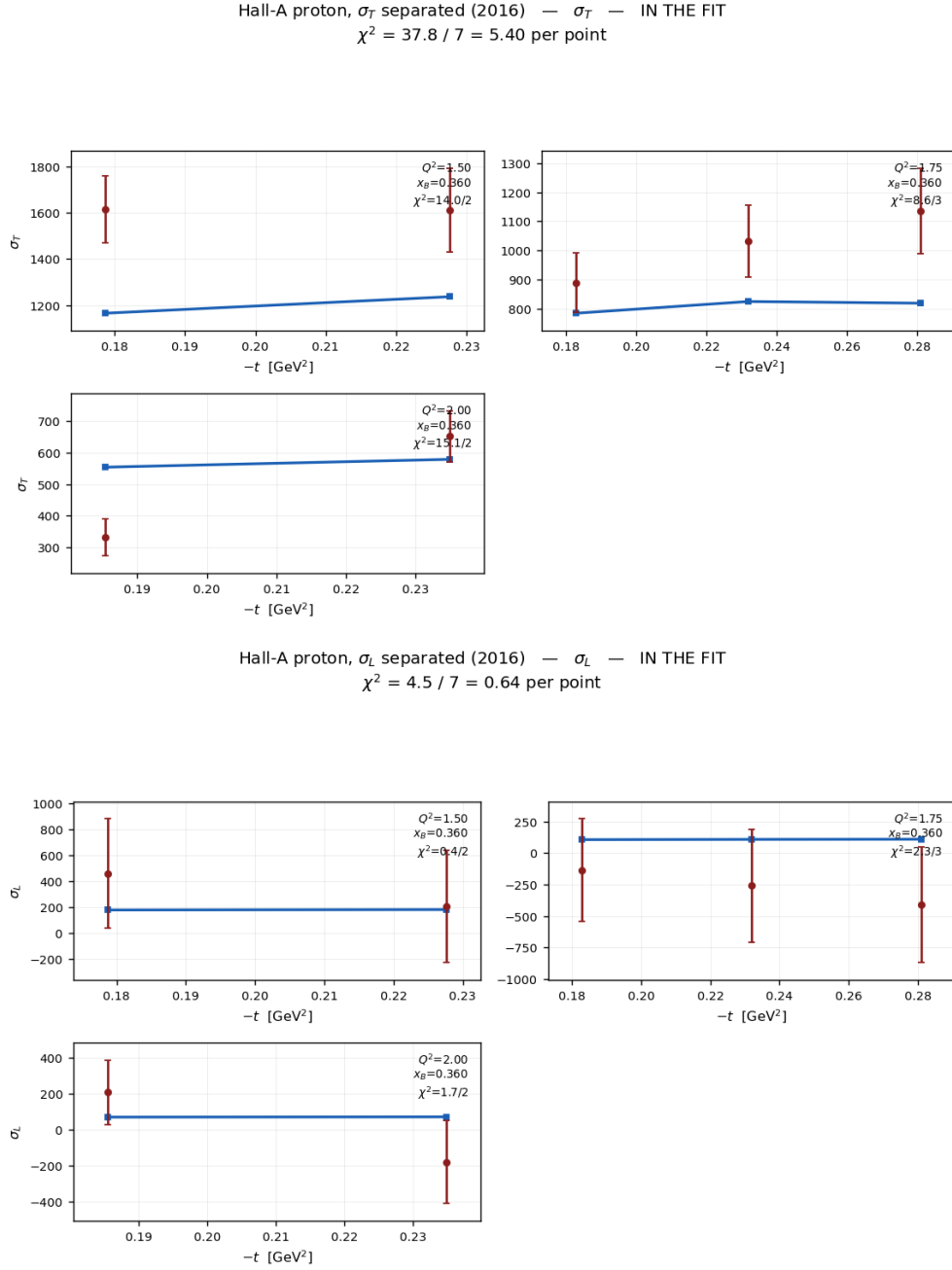


Figure 10: Hall A 2016, the Rosenbluth separation. σ_T above; σ_L below, recovered from the publication and absent from every fit before October 2026. Note the vertical scale of the lower panel: the errors are of the size of the values.

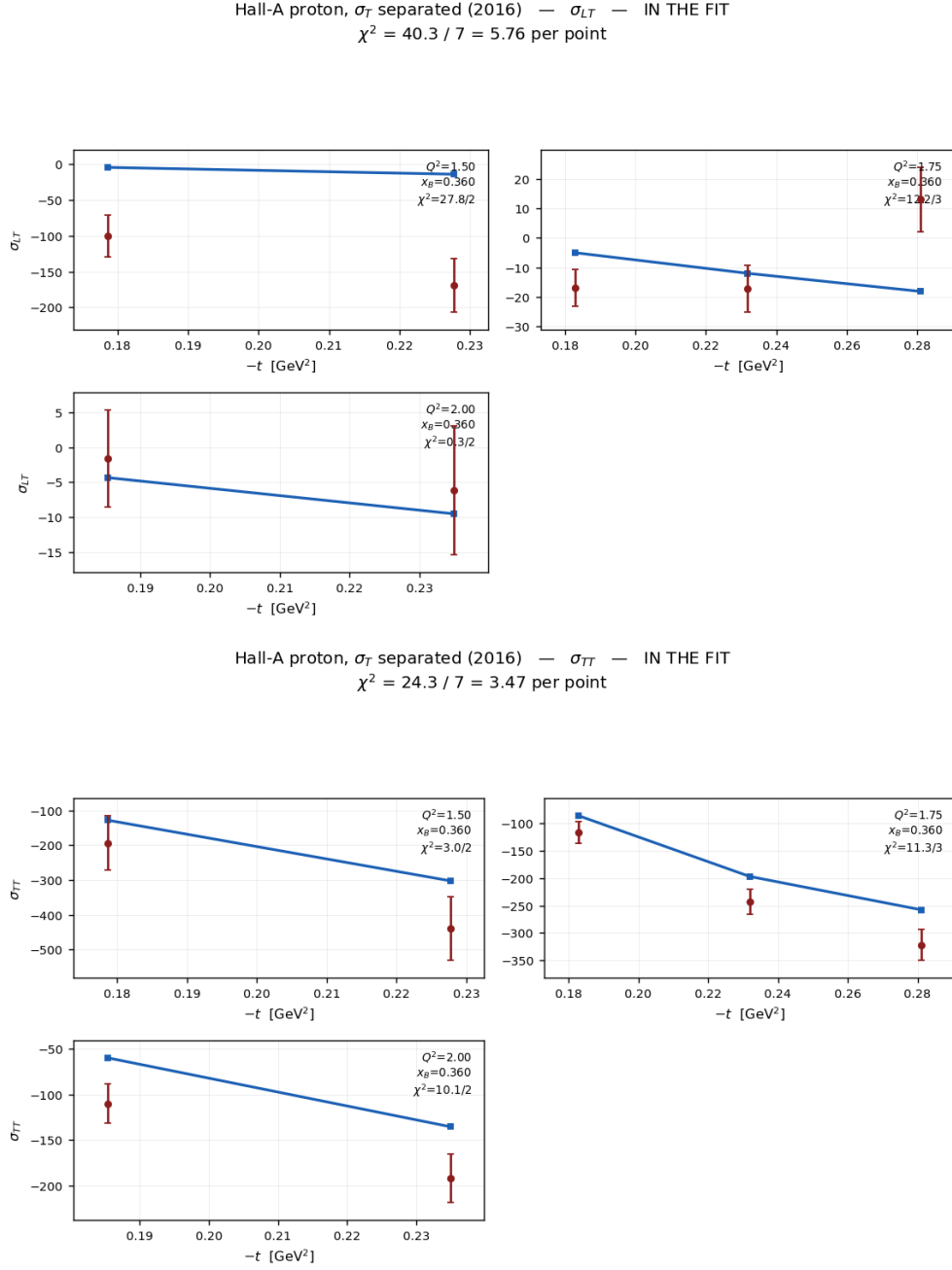


Figure 11: Hall A 2016, σ_{LT} and σ_{TT} . The two $Q^2 = 1.50$ points of the upper panel are two of the three that no smooth model reaches, section 5.

Hall-A proton 6 GeV (2011) — $\sigma_U = \sigma_T + \epsilon\sigma_L$ — IN THE FIT
 $\chi^2 = 39.5 / 28 = 1.41$ per point

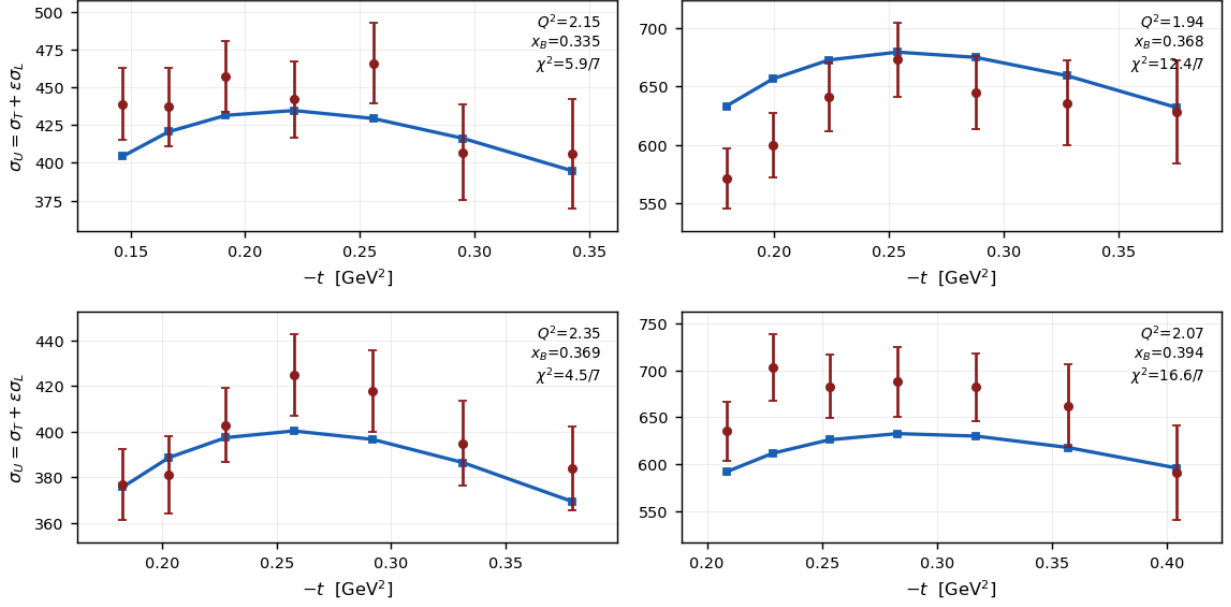


Figure 12: Hall A 2011, σ_U .

Hall-A proton 6 GeV (2011) — σ_{LT} — IN THE FIT
 $\chi^2 = 45.0 / 28 = 1.61$ per point

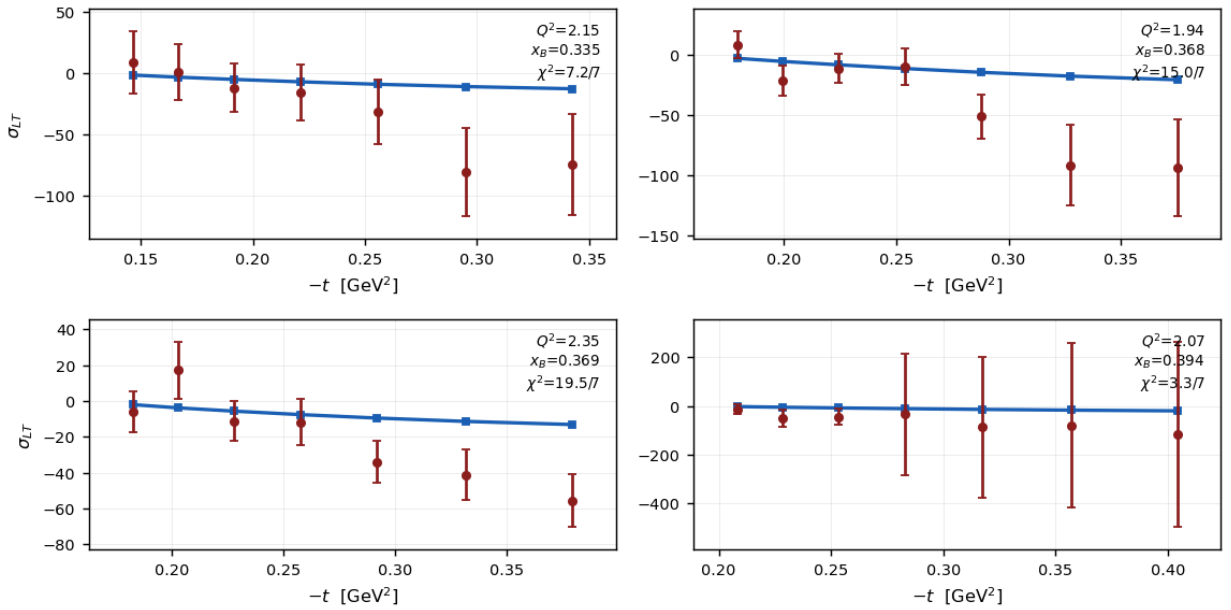


Figure 13: Hall A 2011, σ_{LT} .

Hall-A proton 6 GeV (2011) — σ_{TT} — IN THE FIT
 $\chi^2 = 34.0 / 28 = 1.21$ per point

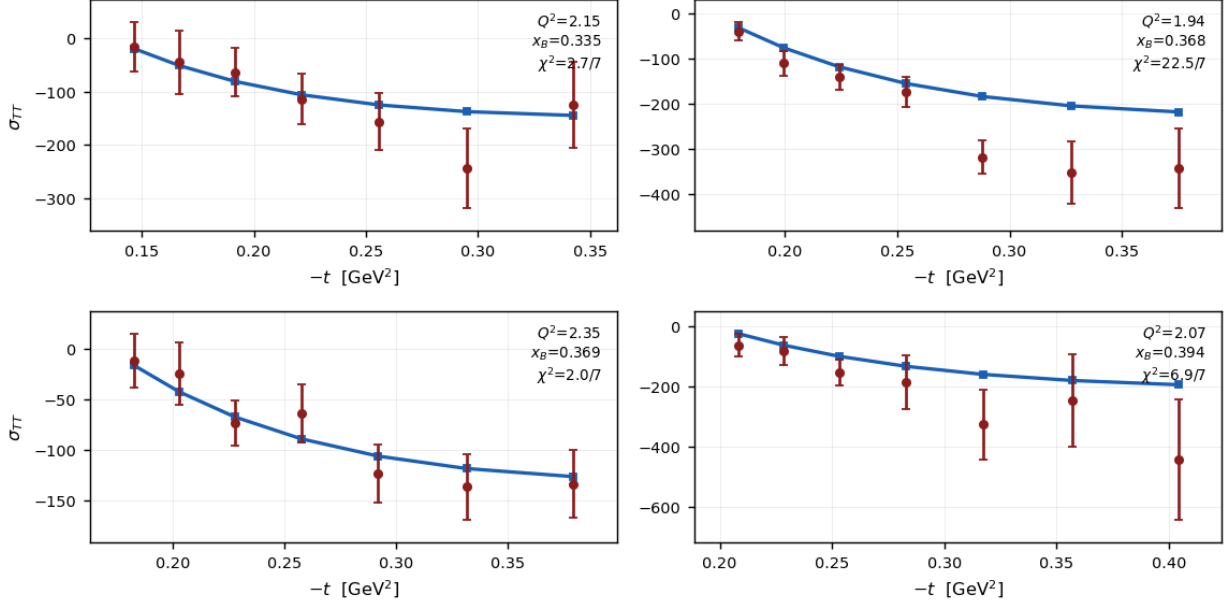


Figure 14: Hall A 2011, σ_{TT} .

Hall-A proton 6 GeV (2011) — $\sigma_{LT'}$ — IN THE FIT
 $\chi^2 = 15.6 / 28 = 0.56$ per point

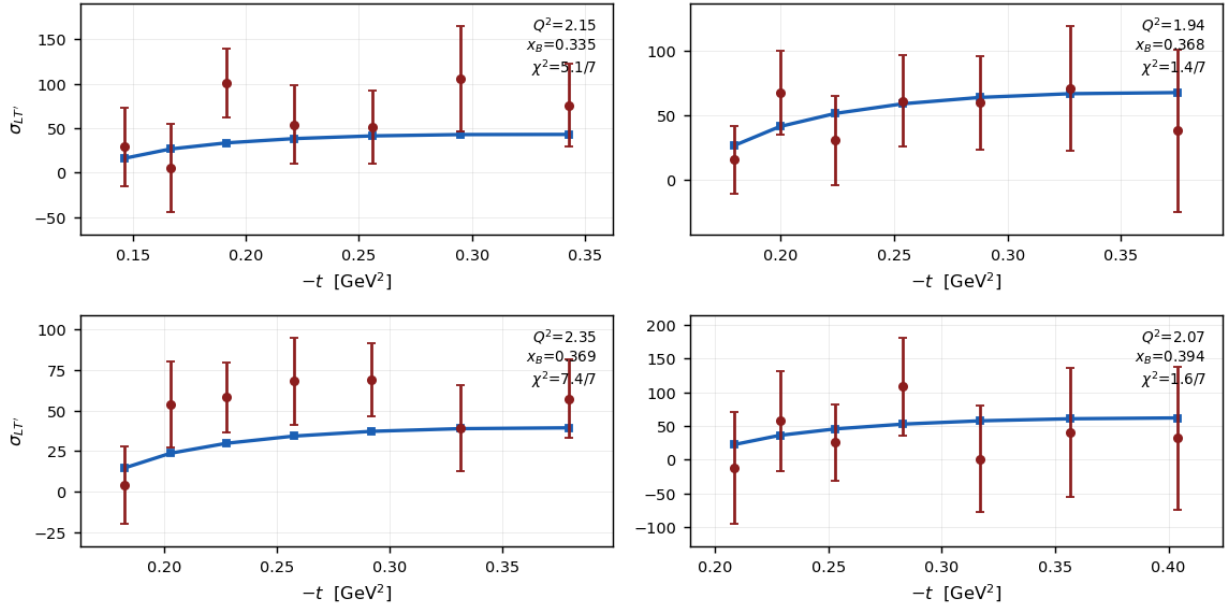


Figure 15: Hall A 2011, $\sigma_{LT'}$ — one of the two direct measurements of the quantity that section 7 finds too small in the model.

Hall-A proton 12 GeV (2021) — $\sigma_U = \sigma_T + \varepsilon \sigma_L$ — IN THE FIT
 $\chi^2 = 105.0 / 36 = 2.92$ per point

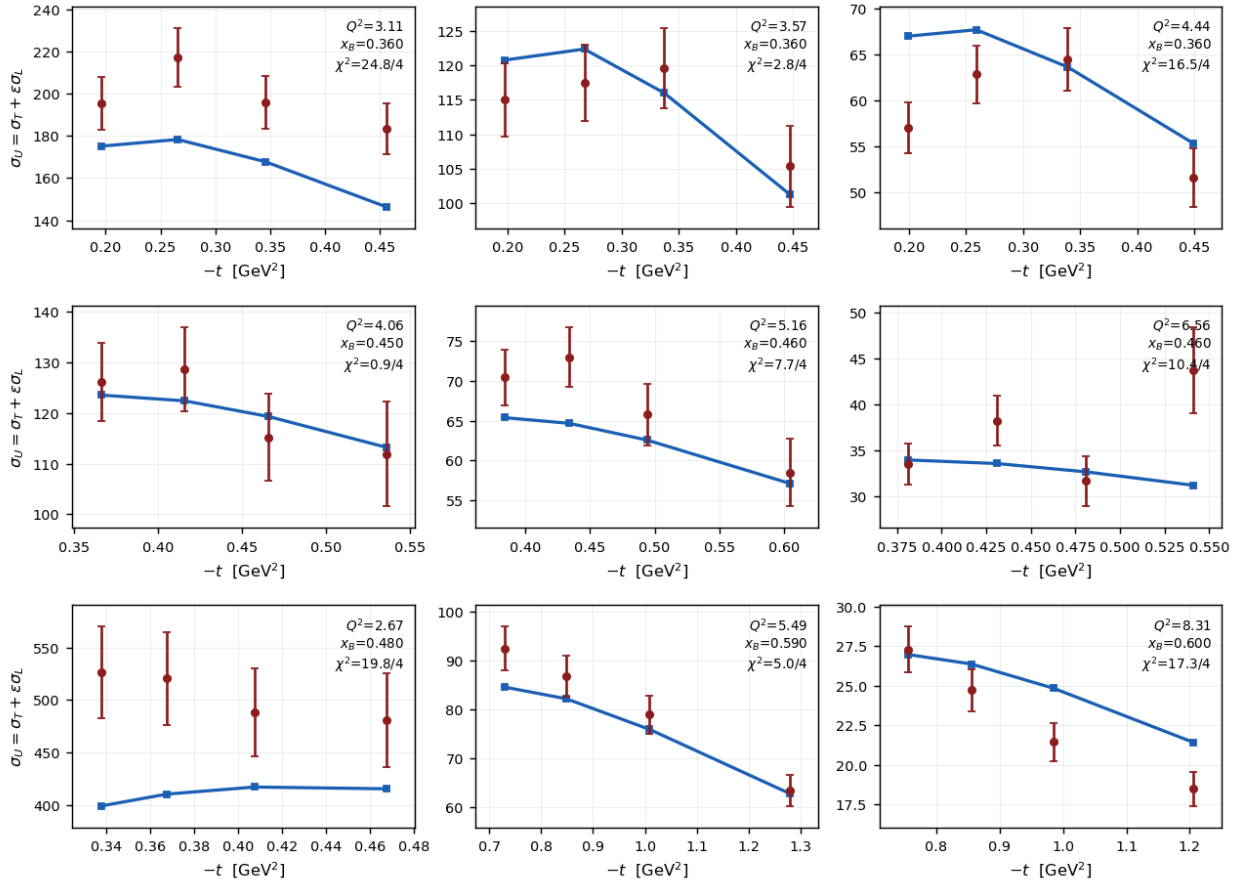


Figure 16: Hall A 2021, σ_U .

Hall-A proton 12 GeV (2021) — σ_{LT} — IN THE FIT
 $\chi^2 = 85.2 / 36 = 2.37$ per point

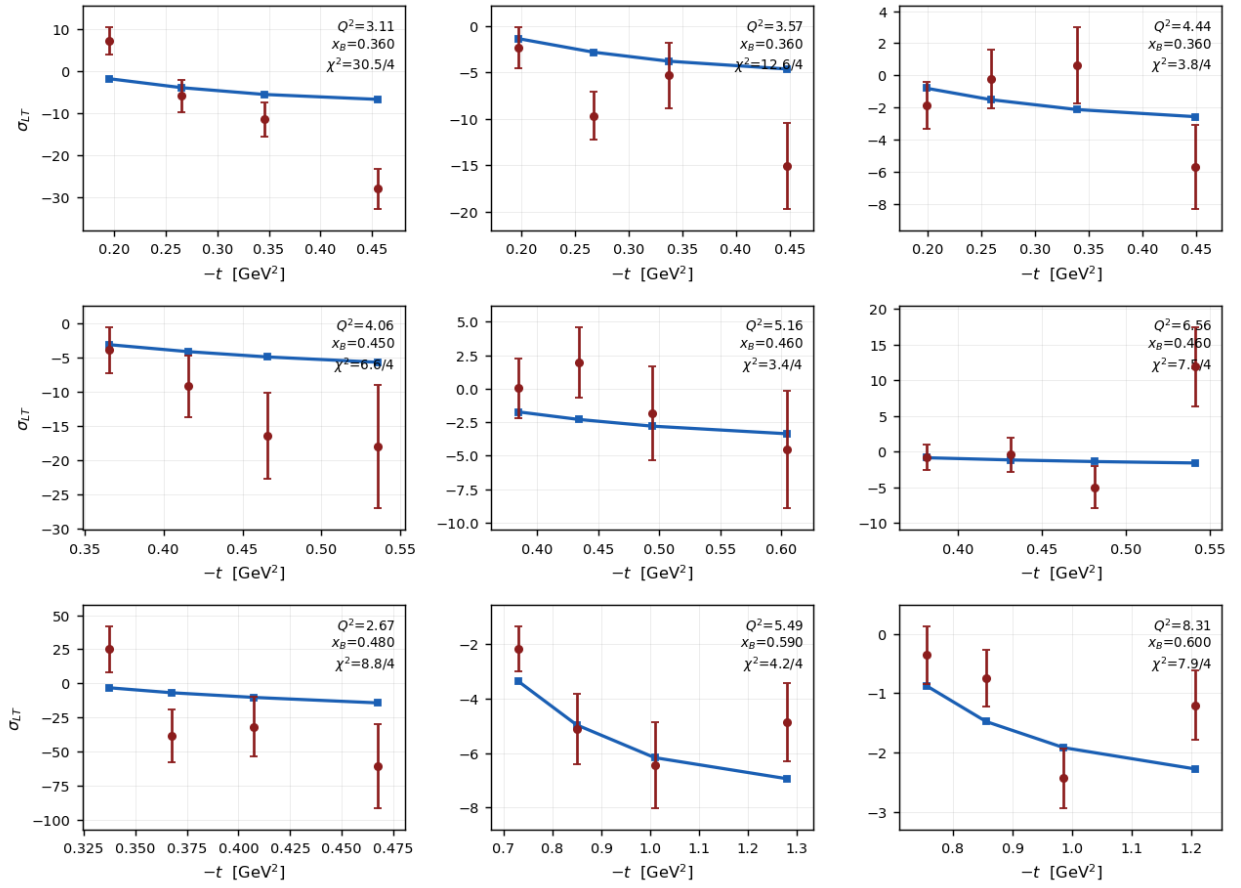


Figure 17: Hall A 2021, σ_{LT} .

Hall-A proton 12 GeV (2021) — σ_{TT} — IN THE FIT
 $\chi^2 = 108.0 / 36 = 3.00$ per point

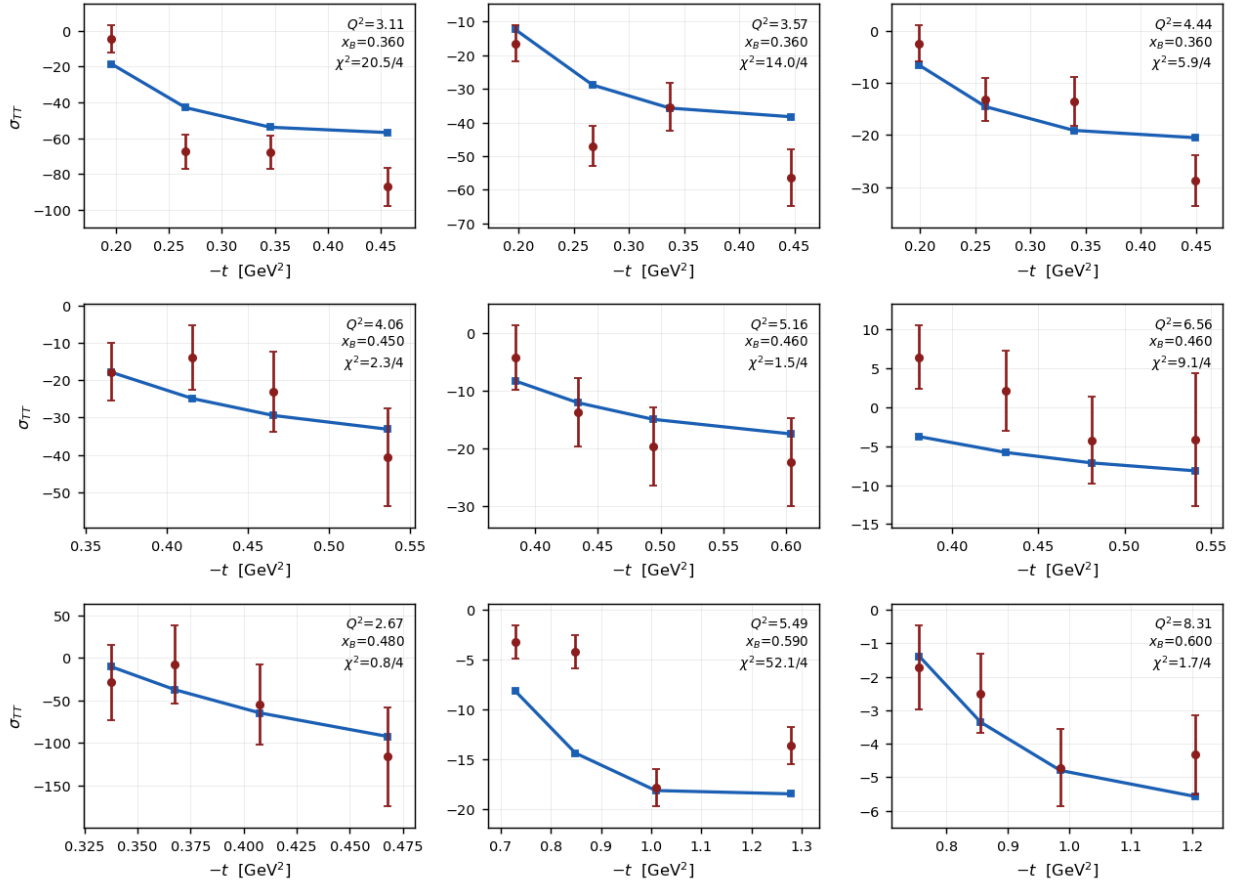


Figure 18: Hall A 2021, σ_{TT} .

Hall-A proton 12 GeV (2021) — $\sigma_{LT'}$ — IN THE FIT
 $\chi^2 = 48.4 / 36 = 1.35$ per point

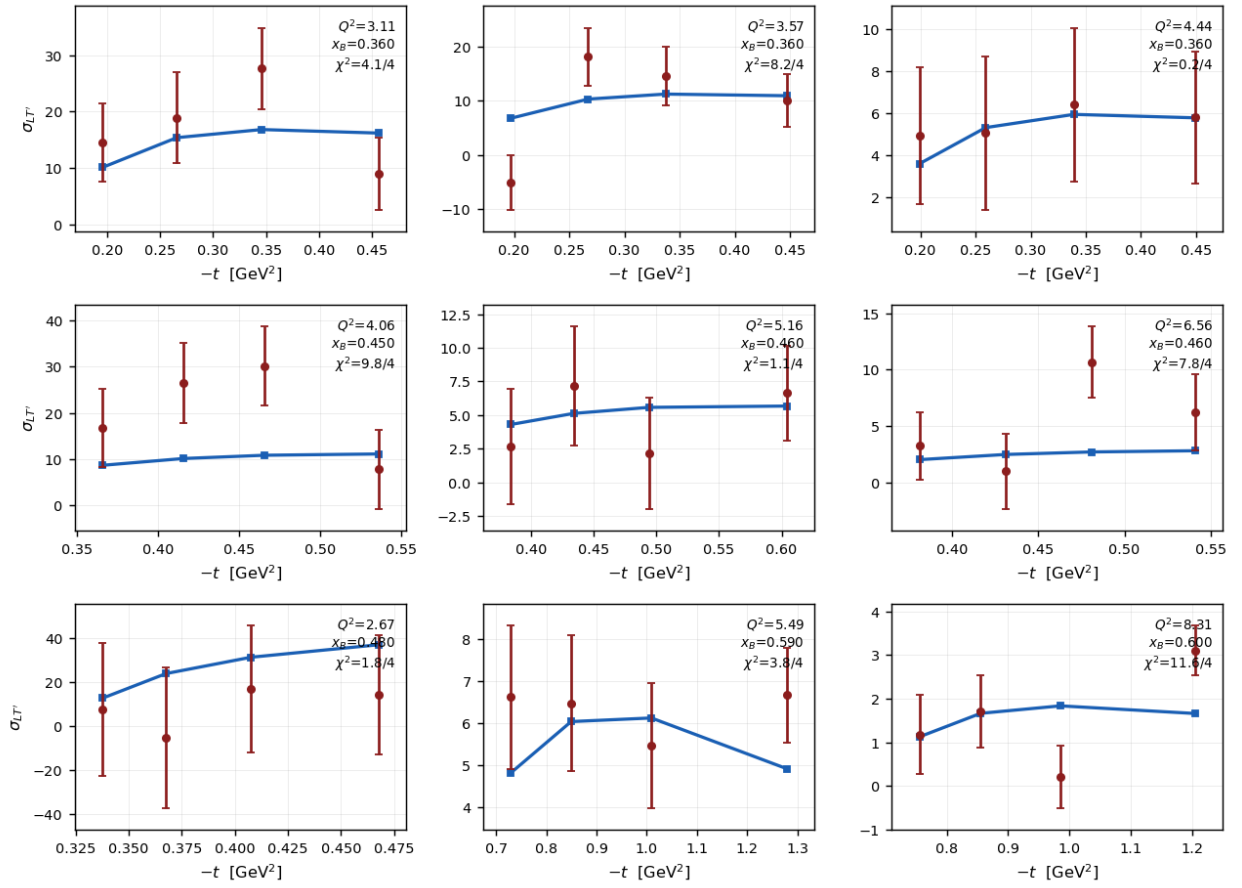


Figure 19: Hall A 2021, $\sigma_{LT'}$.

CLAS12 cross sections, preliminary — $\sigma_U = \sigma_T + \epsilon\sigma_L$ — NOT in the fit (blind prediction)
 $\chi^2 = 411.7 / 146 = 2.82$ per point

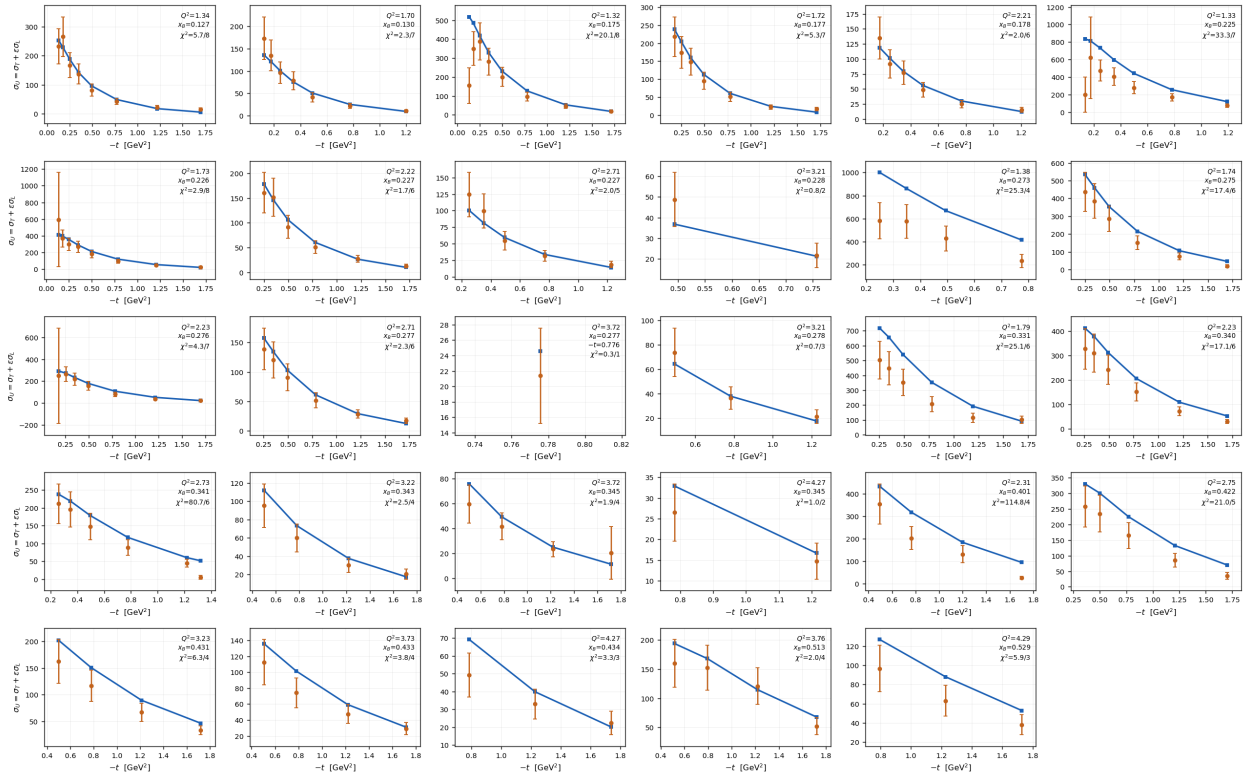


Figure 20: CLAS12 cross sections, σ_U — NOT IN THE FIT.

CLAS12 cross sections, preliminary — σ_{LT} — NOT in the fit (blind prediction)
 $\chi^2 = 4685.9 / 146 = 32.10$ per point

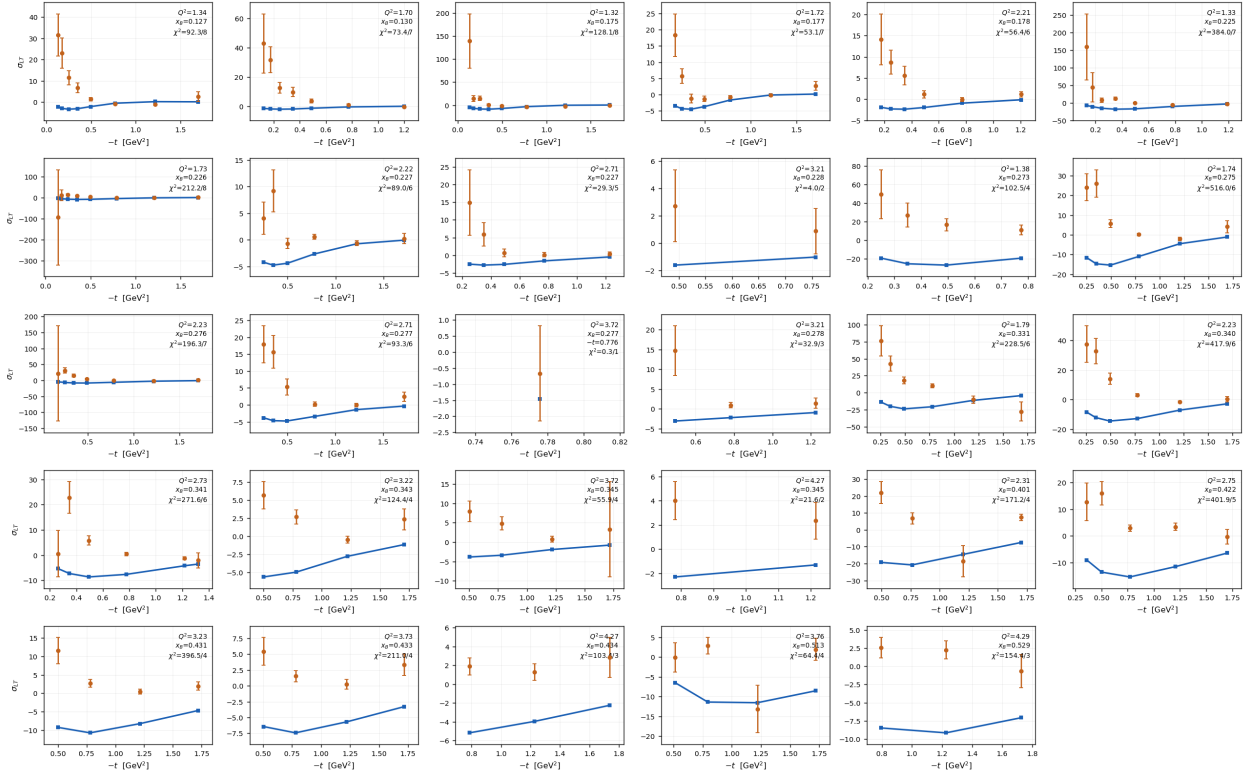


Figure 21: CLAS12, σ_{LT} — NOT IN THE FIT. This is the sign disagreement of section 7.

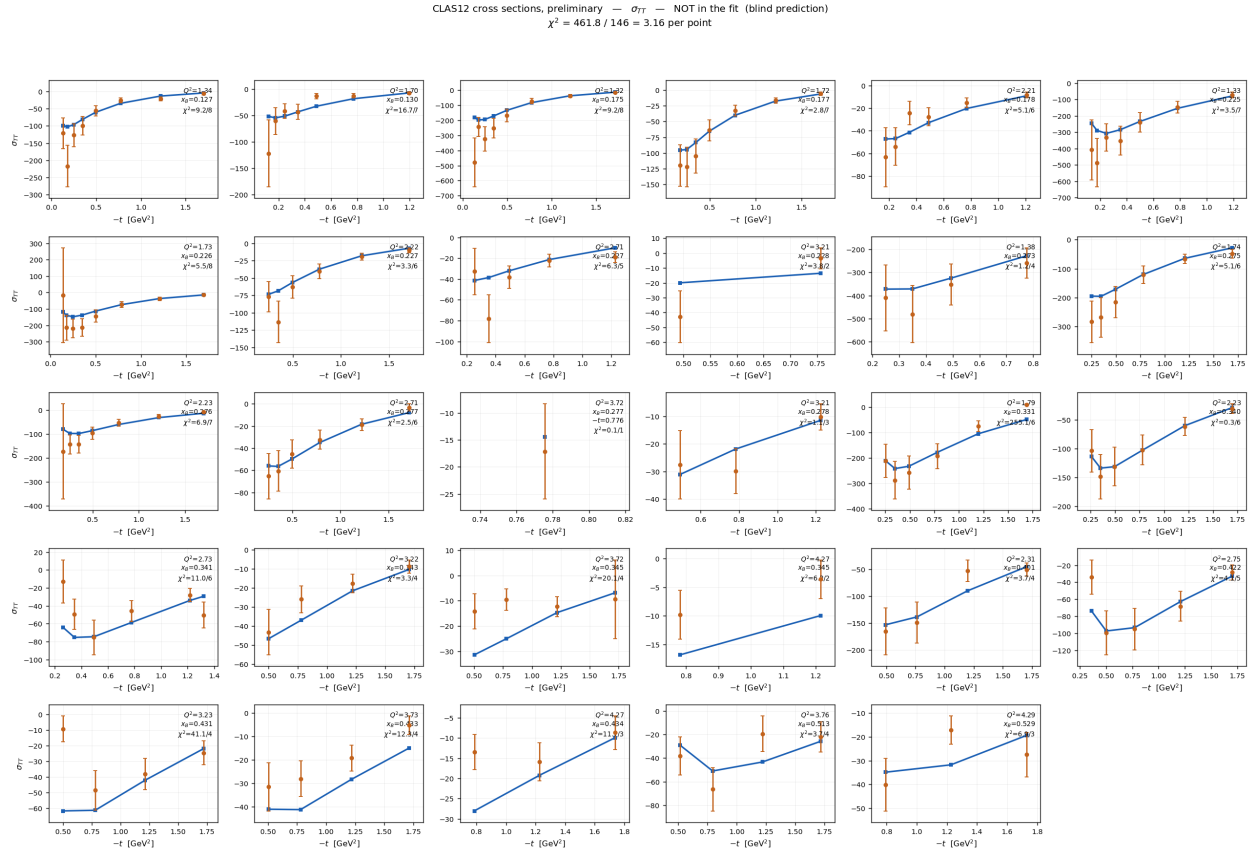


Figure 22: CLAS12, σ_{TT} — NOT IN THE FIT.

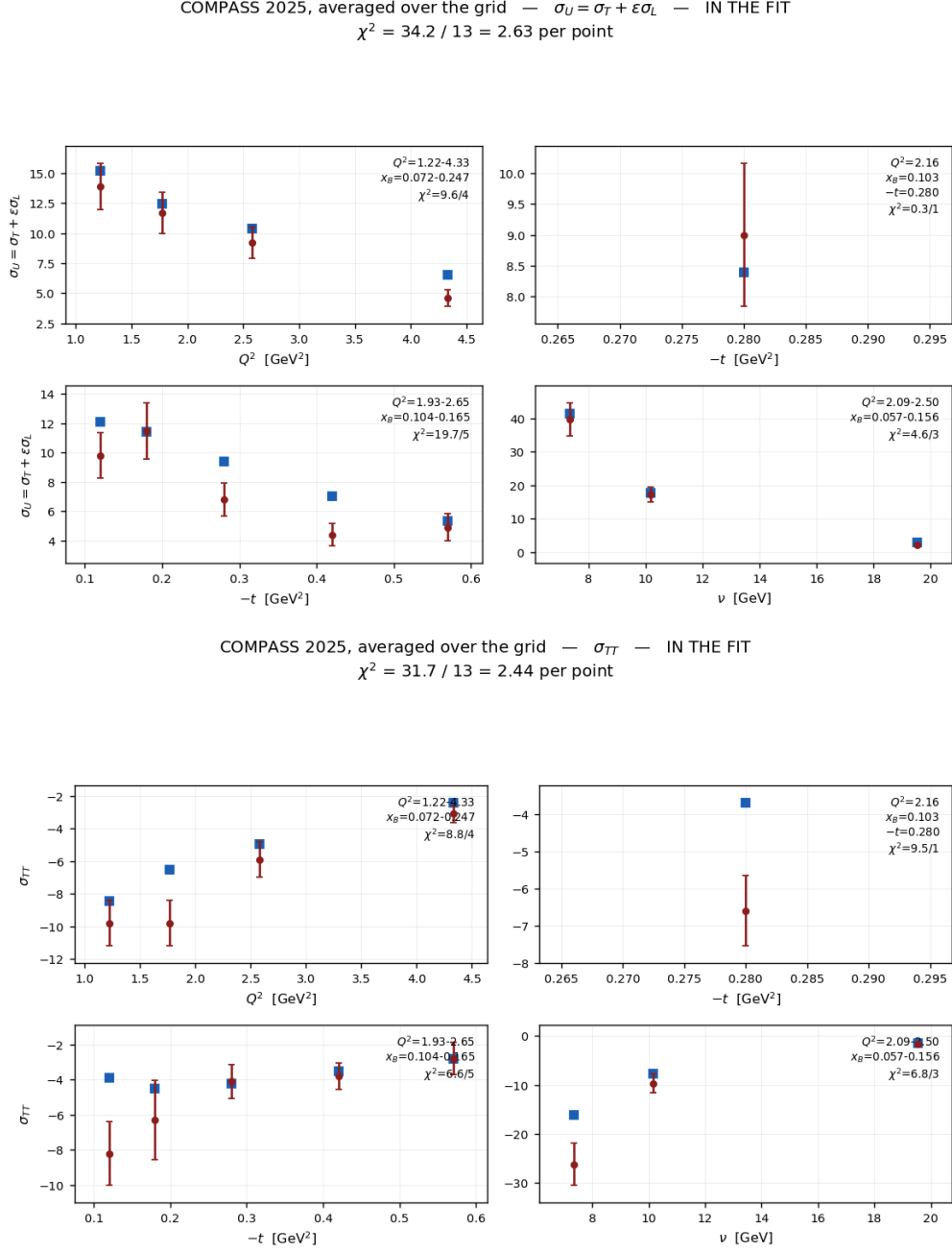


Figure 23: COMPASS, averaged over the published four-dimensional grid rather than evaluated at $\langle Q^2 \rangle, \langle x_B \rangle$: one cell spans x_B from 0.02 to 0.47, and evaluating at the mean overshoots by a factor five to eight.

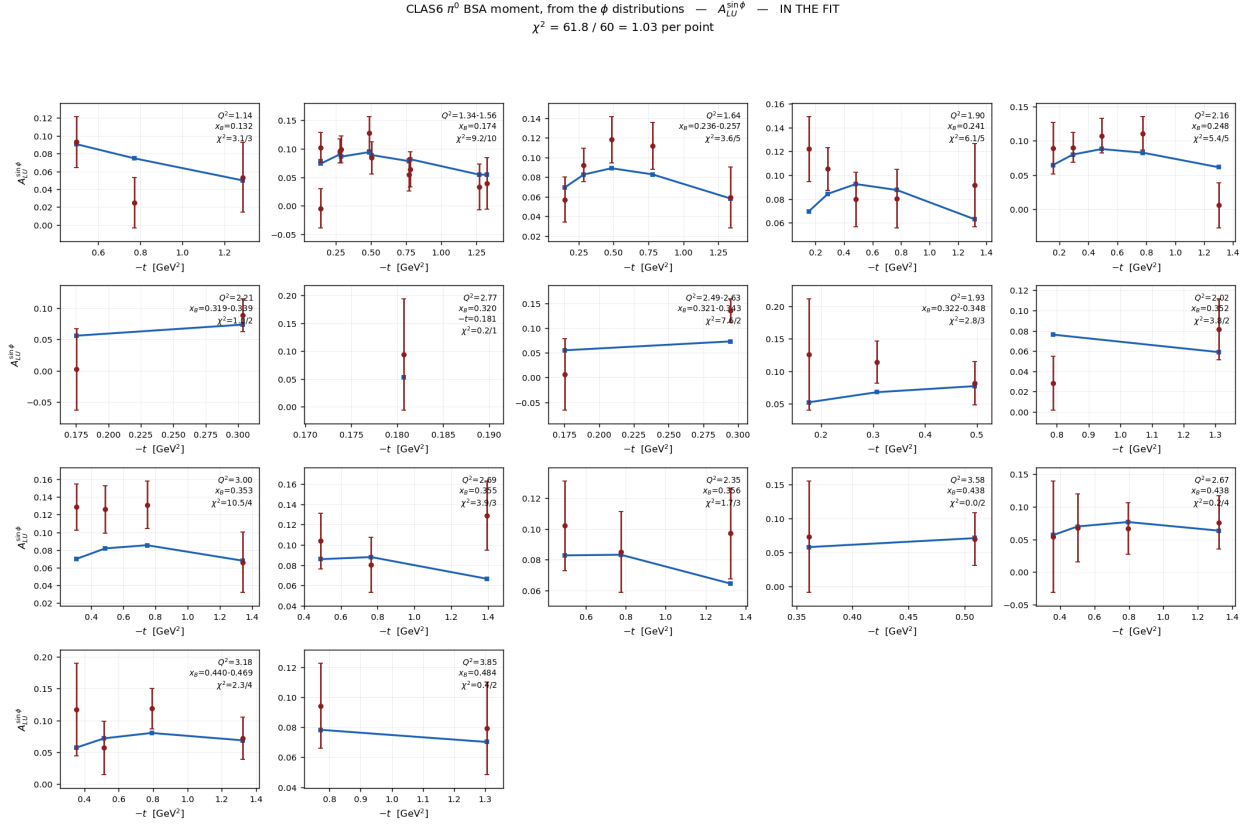


Figure 24: De Masi, the $\sin\phi$ moment extracted from the ϕ distributions with the full asymmetry form, section 2.2.

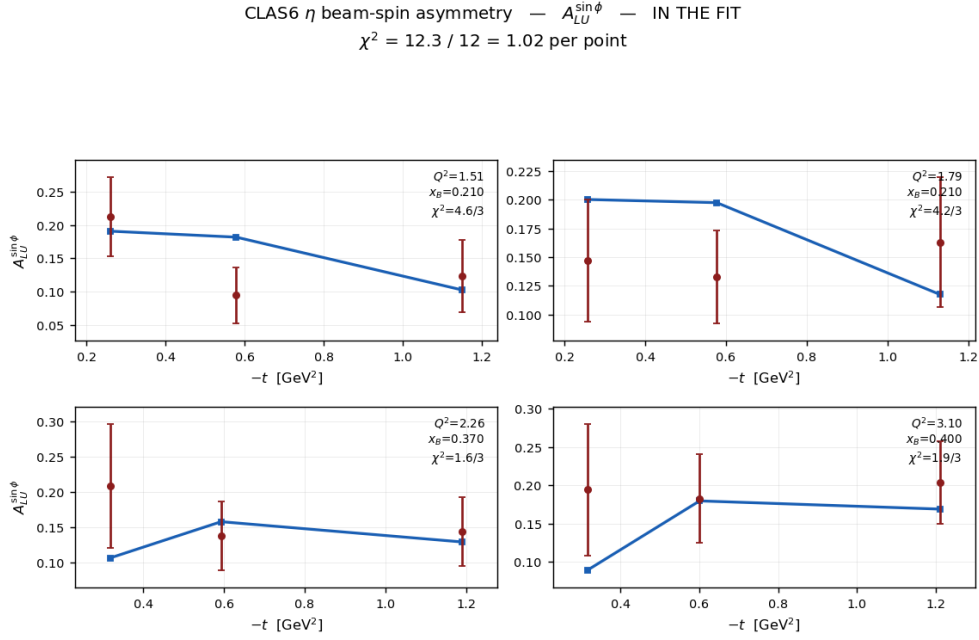


Figure 25: Zhao, the η beam-spin asymmetry.

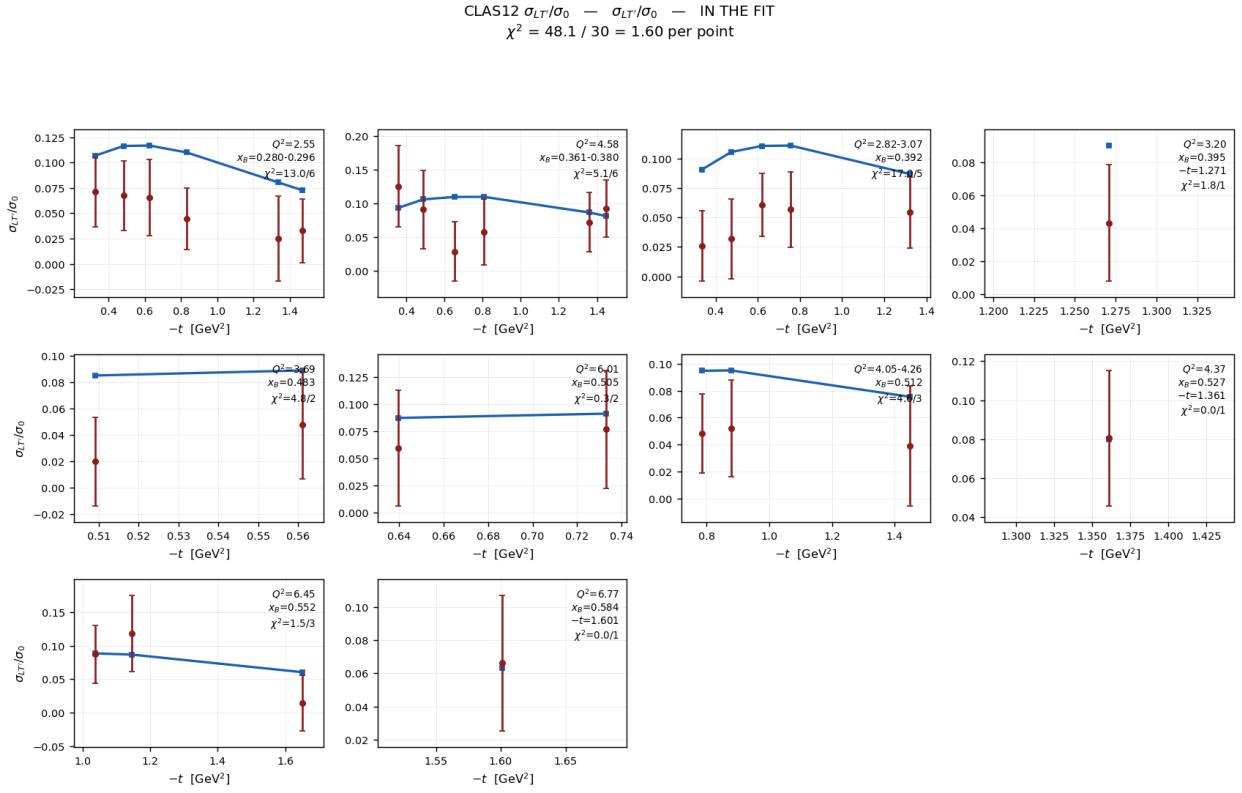


Figure 26: CLAS12 $\sigma_{LT'}/\sigma_0$.

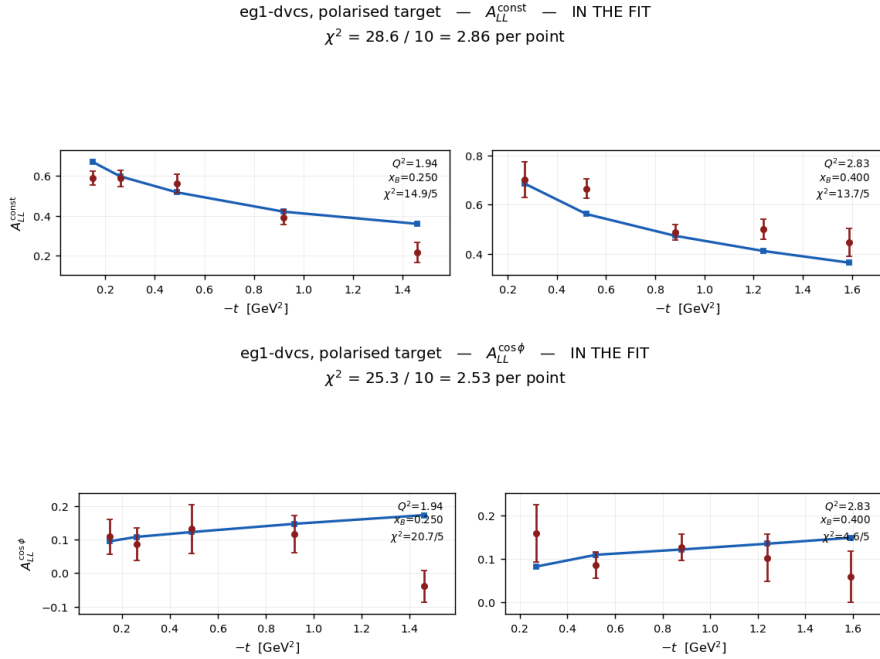
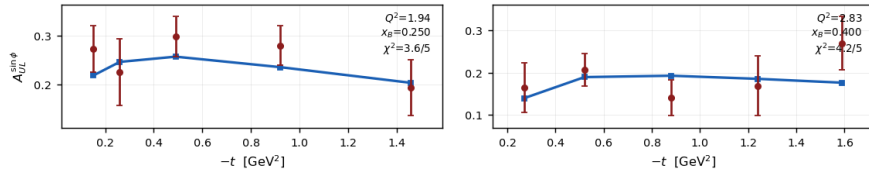


Figure 27: eg1-dvcs, the A_{LL} moments — the only observable in the database that sees the $T_{01}-U_{01}$ relative phase.

eg1-dvcs, polarised target — $A_{UL}^{\sin\phi}$ — IN THE FIT
 $\chi^2 = 7.8 / 10 = 0.78$ per point



eg1-dvcs, polarised target — $A_{UL}^{\sin 2\phi}$ — IN THE FIT
 $\chi^2 = 20.5 / 10 = 2.05$ per point

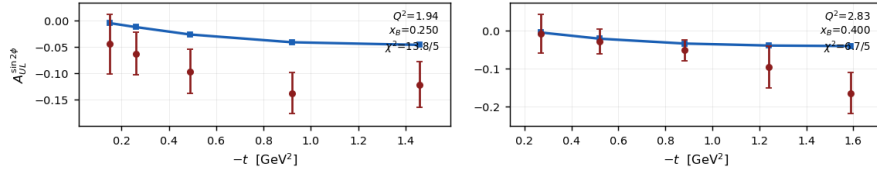


Figure 28: eg1-dvcs, the A_{UL} moments.

Reproducing this

Code, data and the full history are at https://github.com/vkubarovsky/rc_iter1 in the repository `pi0eta_ampmodel_fit`. The fit shown is run `Mom_B12`; its parameter vector is `runs/Mom_B12/fitpar.npy` and every number in the tables above is generated from it by `tex/make_tables.py`, so nothing here is transcribed by hand. The De Masi moments are `data/demasi_moments.txt`, written by `demasi_moments.py`, whose header records the reference model used for the denominator and its md5. The recovered Rosenbluth blocks are in `halla_rosenbluth.py`. The remaining data are `pi0_eta_database.xlsx`; `export_db.py --check` verifies that the CSV cache the fit reads still matches the workbook.